Basic Linear Algebra (융합기초선형대수학)

KNU Math 902

Classnotes

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These notes are based on the **International Student Edition** of the **Fourth Edition** of Gilbert Strang's delightful book **'Linear Algebra and its applications'**. We refer to this as 'The Text'. Be sure to get The Text as we deliberately refer to it often. These notes cannot be used well without it.

1 Matrices and Gaussian Elimination

1.1 Introduction

The basic object of study of this course is a system, or set, of linear equations . Here are some examples. Can you solve them? That is, can you find the values x and y, (and z in the last example) that make the equations true?

Example 1.1.

$$\begin{array}{rcrcr} 2x & - & 2y & = & 2 \\ 2x & + & y & = & 5 \end{array}$$

Example 1.2.

$$\begin{array}{rcrcr} 2x & - & 2y & = & 2 \\ x & - & y & = & 4 \end{array}$$

Example 1.3.

$$\begin{array}{rcrcr} 2x & - & 2y & = & 2 \\ x & - & y & = & 1 \end{array}$$

Example 1.4.

$$\begin{array}{rcrcrcr} x & + & 2y & - & 3z & = & 3 \\ x & - & 2y & + & 5z & = & 7 \\ x & + & y & + & 7z & = & 9 \end{array}$$

In this course we look at

- deciding if a system of equations has a solution,
- solving systems of equations,
- solving them quickly,
- approximating solutions (more quickly), and
- interpreting solutions geometrically.

In all of these tasks, it will be convenient to represent systems of linear equations with matrices. However, we put the definition of a matrix off for a while, letting our interpretations of systems or equations suggest it naturally.

linear: 선형 (적) equation: 방정식 solution: 해결책 approximate: 대략 geometry: 기학 matrix: 행렬

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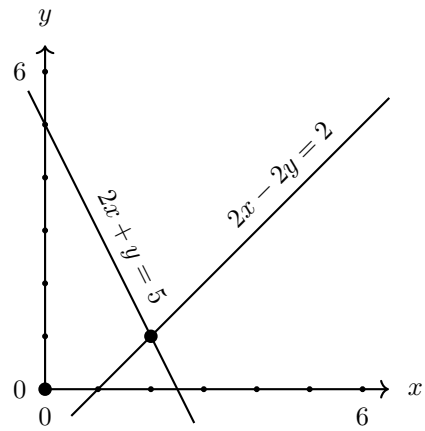
1.2 The Geometry of Linear Equations

There are two basic ways to interpret a system of linear equations geometrically.

The row picture

Each row is an equation that we can graph. The rows of the system in Example 1.1 can be graphed as follows. The solution $(x, y) = (2, 1)$ is the intersection of the two lines.



Practice

Draw similar pictures for Examples 1.2 and 1.3. What do 'no solutions' or 'many solutions' correspond to under this point of view? How is the picture different for Example 1.4?

The column picture

For the column view we need some notation. An *n-dimensional vector* (over $\mathbb{R}$) is an ordered set of n real numbers: $(1, 2, 1/2)$ and $(1/2, 2, 1)$ are different 3-dimensional vectors. They are *row vectors* but we could write them as column vectors:

$$\begin{bmatrix} 1 \\ 2 \\ \frac{1}{2} \end{bmatrix} \text{ and } \begin{bmatrix} \frac{1}{2} \\ 2 \\ 1 \end{bmatrix}.$$

Any n-dimensional vector can be interpreted as a point in $\mathbb{R}^n$, or as the arrow from the origin to that point, depending on what we want to do with it.

Example 1.5. The vector $(1, 2, 2)$ can be viewed either as the point or the arrow in the following picture.

row: 행 intersection: 교차점 column: 렬 dimension: 차원

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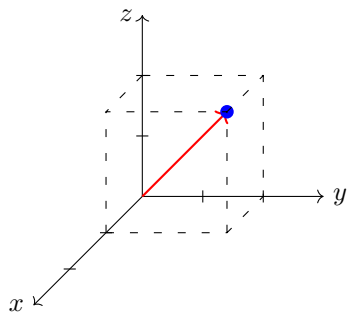
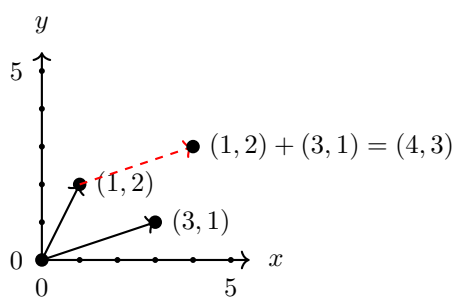


We add vectors componentwise:

$$\begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}3\\1\end{bmatrix} = \begin{bmatrix}4\\3\end{bmatrix}$$

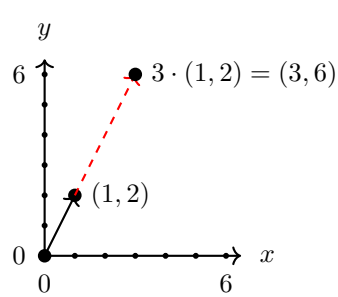
which corresponds to concatenating arrows 'head-to-tail'



and we can multiply them by a scalar:

$$3\begin{bmatrix}1\\2\end{bmatrix} = \begin{bmatrix}3\\6\end{bmatrix}$$

which corredsponds to stretching or shrinking or flipping them.



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The system of equations from Example 1.1

$$\begin{array}{rcrcr} 2x & - & 2y & = & 2 \\ 2x & + & y & = & 5 \end{array}$$

can be represented as an equation of column vectors

$$x\begin{bmatrix}2\\2\end{bmatrix} + y\begin{bmatrix}-2\\1\end{bmatrix} = \begin{bmatrix}2\\5\end{bmatrix}.$$

Practice

Try to intepret a solution to this equation of column vectors in terms of the column picture.

Problems from the text

Sect 1.2: 2,5,9,16 (Non International Student Edition: 4,3,6,16)

1.3 Gaussian Elimination

Let's now find the solutions to the system of equations in Example 1.4; we label the individual equations as R_i and refer to them as rows.

$$\begin{array}{rcrcrcrcr} R_1: & x & + & 2y & - & 3z & = & 3 \\ R_2: & x & - & 2y & + & 5z & = & 7 \\ R_3: & x & + & y & + & 7z & = & 9. \end{array}$$

To solve the system we try to do a series of *row operations* of the following types:

- *elimination*: $R_i := R_i + cR_j$ for $j < i$. Add c times the jth row to the ith row.
- *back-substitution*: Same as elimination, but $i < j$.
- *scaling*: $R_i := cR_i$. (Multiply the ith row by a non-zero scalar c.)
- *row switching*: $R_i \leftrightarrow R_j$. Switch the ith and j^{th} rows.

Applying a row operation to a system of equations does not change its solutions.

Practice

Show this. Hint: if x, y, and z are such that $x + 2y - 3z = 3$ and $x - 2y + 5z = 7$ are true, then why is, say, $R_2 = R_2 + R_1 : 2x + 2z = 10$ true?

Our goal is to get to a system that looks like the following.

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$$\begin{array}{rrrcr} 1x & & & = & 4\frac{3}{8} \\ & -4y & & = & -1 \\ & & 8z & = & 5 \end{array}$$

The coefficient of the ith variable in the ith equation is called the ith *pivot*. (We will properly define 'pivot' at the end of the section, after we have seen some more examples.) *Gaussian Elimination* does this in two main steps.

i. Elimination: For $i = 1$ to $n-1$, clear all entries below the ith pivot using elimination operations.

ii. Back substitution: For $i = n$ to 2, clear all entries above the ith pivot using back-substitution operations.

If all goes right, the elimination step *triangulates* the system:

$$\begin{bmatrix} x + 2y - 3z = 3 \\ x - 2y + 5z = 7 \\ x + y + 7z = 9 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} x + 2y - 3z = 3 \\ -4y + 8z = 4 \\ x + y + 7z = 9 \end{bmatrix} \xrightarrow{R_3 - R_1} \begin{bmatrix} x + 2y - 3z = 3 \\ -4y + 8z = 4 \\ -1y + 10z = 6 \end{bmatrix} \xrightarrow{R_3 - \frac{1}{4}R_2} \begin{bmatrix} x + 2y - 3z = 3 \\ -4y + 8z = 4 \\ 8z = 5 \end{bmatrix}$$

This is called the *row echelon form* From here we can solve the system: solving $z = 5/8$ then plugging that into the second row and getting $-4y + 8(5/8) = 4$ which we solve to get $y = 1/4$, and the continuing up. This is back substitution, and we will continue doing it with back-substitution operations to *diagonalise* the system.

$$\begin{bmatrix} x + 2y - 3z = 3 \\ -4y + 8z = 4 \\ 8z = 5 \end{bmatrix} \to \cdots \to \begin{bmatrix} x + 2y = 4\frac{7}{8} \\ -4y = -1 \\ 8z = 5 \end{bmatrix} \to \begin{bmatrix} x = 4\frac{3}{8} \\ -4y = -1 \\ 8z = 5 \end{bmatrix}$$

Solving now is even easier. Our solution is $(x, y, z) = \frac{1}{8}(35, 2, 5)$. We have three pivots: $1, -4$ and 8. Until now, we have only used elimination operations and back substitution operations. Often the calculations can be simplified by using row scalings and row switches, but this is not, strictly, Gaussian Elimination. One could continue from here with some row scalings to get the system

$$\begin{bmatrix} x & & & = & 4\frac{3}{8} \\ & y & & = & 1/4 \\ & & z & = & 5/8 \end{bmatrix}$$

from which it is even easier to read off the solution. But we will see that there is some significance to using only elimination and back-substitution operations if we can. For one, though we still know that we had three non-zero pivots, we no longer know what they are.

Now, all this is a heck of a lot of writing. Let's use some shorthand. We almost did it above. Let's just remove the variables and replace the equals signs with a big line. Writing

$$\begin{array}{rcrcrcr} x & + & 2y & - & 3z & = & 3 \\ x & - & 2y & + & 5z & = & 7 \\ x & + & y & + & 7z & = & 9 \end{array} \qquad \text{as} \qquad \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 1 & -2 & 5 & 7 \\ 1 & 1 & 7 & 9 \end{array}\right]$$

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elimination becomes

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 1 & -2 & 5 & 7 \\ 1 & 1 & 7 & 9 \end{array}\right] \xrightarrow{R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 3 \\ 0 & -4 & 8 & 4 \\ 1 & 1 & 7 & 9 \end{array}\right] \to \dots.$$

Practice

Solve the following systems. Try to eliminate them using only elimination operations, and then to back substitute using only back-substitution operations. You might encounter some problems. How should you deal with them?

Example 1.6.

$$\begin{array}{rcrcrcr} 2u & + & v & + & w & = & 5 \\ 4u & - & 6v & & & = & -2 \\ -2u & + & 7v & + & 2w & = & 9 \end{array}$$

Example 1.7.

$$\begin{array}{rcrcrcr} x & + & 2y & + & 2z & = & 4 \\ 2x & + & 4y & + & z & = & 3 \\ x & + & 3y & + & z & = & 1 \end{array}$$

Example 1.8.

$$\begin{array}{rcrcrcr} x & + & 2y & + & 2z & = & 4 \\ x & + & 2y & + & 3z & = & 4 \\ 2x & + & 4y & + & z & = & 2 \end{array}$$

Example 1.9.

$$\begin{array}{rcrcrcr} x & + & 2y & + & 2z & = & 4 \\ x & + & 3y & + & 3z & = & 4 \\ & & y & + & z & = & 0 \end{array}$$

We saw two things: to eliminate a system we might need to use more than elimination operations. We had to use a row switch, or possibly some back-substitution operations in the elimination step. We also saw that a system might not be diagonalisable. In this case our pivots were not on the diagonal. So what are pivots? Let's clarify this.

When doing Gaussian elimination to a matrix to reduce it to an upper triangular matrix, using only elimimation row operations, the ith *pivot* is the first non-zero co-efficient in the ith row.

Problems from the text

1.3: 1, 11, 12, 15, 22 (5, 12, 10, 18, 20)

1.4 Matrices

We 'kind of' used a matrix in the last section, but now we define it properly.

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An m by n matrix is a list of $m \times n$ numbers arranged in m rows of n numbers each. For example,

$$\begin{bmatrix} 1 & 2 & -2 & 4 \\ 1 & 1/3 & 3 & \pi \\ \sin(1/2) & 1 & 1 & 0 \end{bmatrix}$$

is a 3×4 matrix.

Notation

Given an $m \times n$ matrix A we write $A = [a_{ij}]$ to mean that a_{ij} is the entry in the ith row and the jth column. So $a_{12} = 3$ if

$$[a_{ij}] = \begin{bmatrix} 0 & 3 & 1 \\ 4 & 2 & 5 \end{bmatrix}.$$

Some Special Kinds of Matrices

The pair 'm by n' is the *dimension* of the matrix. A $1 \times n$ matrix is a *column matrix*, and an $m \times 1$ matrix is a *row matrix*. An $n \times n$ matrix is a *square*. If a square matrix $A = [a_{ij}]$ has $a_{ij} = 0$ whenever $i > j$ then it is *upper-triangular* and if $a_{ij} = 0$ whenever $i < j$, it is *lower-triangular*. It it is upper and lower triangular, then it is diagonal.

$$\text{upper-triangular: } \begin{bmatrix} 1 & 2 & 4 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{bmatrix} \qquad \text{lower-triangular: } \begin{bmatrix} 1 & 0 & 0 \\ 3 & 3 & 0 \\ 4 & 7 & 2 \end{bmatrix} \qquad \text{diagonal: } \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

The diagonal matrix with ones all along the diagonal is the *identity:*

$$I = I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

Notationally, matrices will be denoted by capital letters such as A, B and M except that the row and column vectors are often denoted by small bold letters such as $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{x}$.

Addition of Matrices

We add matrices of the same dimensions by adding entrywise:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 5 & 6 \\ 5 & 7 \end{bmatrix} = \begin{bmatrix} 1+5 & 2+6 \\ 3+5 & 4+7 \end{bmatrix}.$$

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Scalar Multiplication

Multiplying a matrix A by a scalar r is in the obvious way: $r[a_{ij}] = [ra_{ij}]$. For example

$$2 \cdot \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 4 & 6 \end{bmatrix}.$$

Matrix Multiplication

This is the more complicated operation. Given an $\ell \times m$ matrix $A = [a_{ij}]$ and an $m \times n$ matrix $B = [b_{ij}]$, the *product matrix* AB is the $\ell \times n$ matrix $C = [c_{ij}]$ where

$$c_{ij} = \sum_{\alpha=1}^{m} a_{i\alpha} \cdot b_{\alpha j}.$$

For example we can take the product of a row and a column vector of the same length:

$$\begin{bmatrix} 1 & 4 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1/2 \end{bmatrix} = [1 \cdot 3 + 4 \cdot 1 + 2 \cdot 1/2] = [8].$$

This is also known as the *dot product* of the two vectors.

Practice

Evaluate

$$\begin{bmatrix} 1 & 7 \\ 2 & 8 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \text{ and } \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 & 3 \\ 2 & 3 & 1 \end{bmatrix}$$

Notice that $AB = BA$ is not generally true, and that the identity is the multiplicative identity.

There are three useful ways to view the product. The first is the *dot product view*. The ij^{th} entry of the product AB is the ith row of A dotted with the jth column of B.

$$\begin{bmatrix} \cdot & \cdot & \cdot \\ \boxed{1 \quad 4 \quad 2} \\ \cdot & \cdot & \cdot \end{bmatrix} \begin{bmatrix} \cdot & \boxed{\begin{matrix} 3 \\ 1 \\ 1/2 \end{matrix}} & \begin{matrix} \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \end{matrix} \end{bmatrix} = \begin{bmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & 8 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \end{bmatrix}$$

The second is the *row view* of AB. We view the rows of A as providing the co-efficeints in linear combinations of the rows of B:

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & 7 \\ 2 & 8 \end{bmatrix} = \begin{bmatrix} 1 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 4 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \\ 2 \cdot \begin{bmatrix} 3 & 7 \end{bmatrix} + 3 \cdot \begin{bmatrix} 2 & 8 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} 11 & 39 \\ 12 & 38 \end{bmatrix}$$

This seems silly perhaps. But we will use it this section in 'encoding' row operations for Gaussian elimination as matrix multiplications.

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The third is the *column view* of AB where we veiw the columns of B a providing co-efficients in linear combinations of the columns of A.

$$\begin{bmatrix}1 & 4\\2 & 3\end{bmatrix}\begin{bmatrix}3 & 7\\2 & 8\end{bmatrix} = \begin{bmatrix}3 & 7\\ \times & \times\\ \begin{bmatrix}1\\2\end{bmatrix} & \begin{bmatrix}1\\2\end{bmatrix}\\ + & +\\ 2 & 8\\ \times & \times\\ \begin{bmatrix}4\\3\end{bmatrix} & \begin{bmatrix}4\\3\end{bmatrix}\end{bmatrix} = \begin{bmatrix}11 & 39\\12 & 38\end{bmatrix}$$

Basic properties of matrix multiplication

We have seen that the identity matrix I is the multiplicative identity, and that matrix multiplication is not generally commuative (ie. $AB = BA$ does not generally hold.) Here are a couple more basic properties.

Fact 1.10. *For any matrices A, B and C of the proper dimensions, the following hold.*

- ***Associativity:*** $(AB)C = A(BC)$
- ***Distributivity:*** $A(B+C) = AB + AC$ *and* $(A+B)C = AC + BC$

As we have an identity matrix, one might ask if there is an inverse? Given a matrix A is there a matrix B such that $AB = I$ or $BA = I$.

Systems of Equations as Matrix equations

Consider the following system of equations.

$$\begin{array}{rcrcl} x & + & 4y & = & 7\\ 2x & - & y & = & 3 \end{array}$$

To solve it, our first step in Gaussian elimination is

$$\left[\begin{array}{cc|c} 1 & 4 & 7\\ 2 & -1 & 3 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 4 & 7\\ 0 & -9 & -11 \end{array}\right]$$

taking 2 times row one from row two. This was shorthand. One can properly write the original system now as

$$\begin{bmatrix}1 & 4\\2 & -1\end{bmatrix}\begin{bmatrix}x\\y\end{bmatrix} = \begin{bmatrix}7\\3\end{bmatrix}.$$

Thinking about the row view of multiplication, the row operation $R_2 - 2R_1$ can by viewed as multiplying on the left by the matrix $\left[\begin{smallmatrix}1 & 0\\-2 & 1\end{smallmatrix}\right]$:

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$$\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 3 \end{bmatrix}.$$

Multiplying this out yields

$$\begin{bmatrix} 1 & 4 \\ 0 & -9 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ -11 \end{bmatrix}$$

which is the system of equations we have after the row operation.

Practice

What matrices do we have to multiply the equation by (on the left) to do a back-substitution operation? A row scaling? A row switch?

The matrices corresponding to row operations (except row switches) are called *elementary matrices*. Except for one entry, they will be 1 on the diagonal, and 0 everywhere else. Those used for elimination will be lower triangular, and those used for back-substitution will be upper triangular.

The matrices corresponding to a row switch are called *permutation matrices*. What do they look like? We will look at them in a couple of sections.

The full Gaussian Elimination (with scaling)

$$\left[\begin{array}{cc|c} 1 & 4 & 7 \\ 2 & -1 & 3 \end{array}\right] \to \cdots \to \left[\begin{array}{cc|c} 1 & 0 & 19/9 \\ 0 & 1 & 11/9 \end{array}\right]$$

can by achieved by multiplying on the left by the matrix:

$$B = \underbrace{\begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}}_{R_1 - 4R_2} \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & -1/9 \end{bmatrix}}_{-R_2/9} \underbrace{\begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}}_{R_2 - 2R_1} = \frac{1}{9} \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix}.$$

Now look. This matrix B is useful. We see, for one:

$$B \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

So thinking of the identity matrix as 1, B is acting like an inverse. Indeed, we define it as this in the next section.

But also, we can use B to solve $\left[\begin{smallmatrix} 1 & 4 \\ 0 & -9 \end{smallmatrix}\right]\left[\begin{smallmatrix} x \\ y \end{smallmatrix}\right] = \left[\begin{smallmatrix} 7 \\ -11 \end{smallmatrix}\right]$, by simply mutiplying it (on the left) on both sides of the equation:

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = B \begin{bmatrix} 1 & 4 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = B \begin{bmatrix} 7 \\ 3 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 19 \\ 11 \end{bmatrix}.$$

Problems from the text

1.4: 2, 4, 7, 11, 13, 19, 26, 43, 44, 45 (3, 7, 2, 10, 17, 21, 22, 40, 41, 46)

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1.6 Inverses and Transposes

Given a matrix A, a matrix L such that $LA = I$ is a *left inverse* of A, and a matrix R such that $AR = I$ is a *right inverse* of A. If they both exist, then we have

$$L = LI = L(AR) = LAR = (LA)R = IR = R,$$

so they are the same. A matrix that is a left and right inverse of A is an *inverse* of A.

Practice

With proofs similar to above, show the following.

i. Only a square matrix can have an inverse.

ii. Not all square matrices A need have a inverse.

iii. If a square matrix A has an inverse, then it is unique.

Definition 1.11. For an $n \times n$ matrix A, the *inverse* A^{-1} is a matrix such that

$$A^{-1}A = I = AA^{-1}.$$

If A has an inverse, it is *invertible* or *non-singular*.

Observe that if two $n \times n$ matrices A and B both have inverses, then so does their product, and $(AB)^{-1} = B^{-1}A^{-1}$. Indeed

$$(AB)B^{-1}A^{-1} = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I.$$

Practice

Thinking of how to express it as a composition of elementary matrices, and inverting them, invert the following upper triangular matrix $\begin{bmatrix} 1 & 3 & 5 \\ 0 & 2 & -3 \\ 0 & 0 & 1 \end{bmatrix}$.

Note

For an $n \times n$ matrix A, we will see that if there is any B such that $BA = I$, then we also have that $AB = I$, and so B is the inverse A^{-1}.

In Section 1.4 we saw that if Gaussian Elimination (with scaling and row switches) can be used to reduce A to I, then it is invertible, and its inverse is the product of the elementary matrices corresponding to the row operations we used in the elimination. We will see that if A is invertible then we can always find the inverse this way. But multiplying out all of these elementary matrices is messy work.

Now, we find the inverse of a matrix in a much neater way, by a process known as Gauss-Jordan elimination.

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Where $A = \left[\begin{smallmatrix} 1 & 4 \\ 2 & -1 \end{smallmatrix}\right]$, construct the augmented matrix

$$[A \mid I] = \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 2 & -1 & 0 & 1 \end{array}\right].$$

This is really just an $n \times 2n$ matrix, but we've added a cosmetic bar down the middle. The idea is that by multiplying by A^{-1} we get $A^{-1}[A \mid I] = [I \mid A^{-1}]$. Multiplying by A^{-1} is really just doing row operations, so if we do these row operations to the augmented matrix, we get A^{-1} in the second part.

Let's try:

$$\left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 2 & -1 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & -9 & -2 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 4 & 1 & 0 \\ 0 & 1 & 2/9 & -1/9 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & 1/9 & 4/9 \\ 0 & 1 & 2/9 & -1/9 \end{array}\right].$$

In three steps we have reduced A to I by multiplying by elementary matrices $E_3E_2E_1$. Their product then is our inverse A^{-1}. In the second part of the augmented matrix, we have multiplied I by the same three elemenatary matrices, so we end up with A^{-1}. Thus $A^{-1} = \frac{1}{9}\left[\begin{smallmatrix} 1 & 4 \\ 2 & -1 \end{smallmatrix}\right]$.

Hey! The inverse is just the original matrix times a constant? Is this always the case?

Practice

Find the inverse of $A = \left[\begin{smallmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 2 & 2 & 3 \end{smallmatrix}\right]$.

Now, we could do almost exactly the same reduction on the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 4 & 3 \\ 2 & 2 & 3 & 4 \end{array}\right]$$

to solve the system of linear equations

$$\begin{array}{ccccccc} x & + & y & + & z & = & 2 \\ 2x & + & 3y & + & 4z & = & 3, \\ 2x & + & 2y & + & 3z & = & 4 \end{array}$$

or equivalently the matrix equation

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 4 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

But we don't have to now that we have A^{-1}. We can use it to 'divide' and so solve the system:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = I \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1}A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}.$$

That is:

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$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A^{-1} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -3 & -1 & 1 \\ 2 & 1 & -2 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} -5 \\ -1 \\ 0 \end{bmatrix}.$$

What's really nice about this is that if we have to solve $A\mathbf{x} = \mathbf{a}$ for several different $\mathbf{a}$, we only have to do the elimination once. For each $\mathbf{a}$ we get away with multiplying a matrix by a vector.

Let's discuss the time this takes in a bit more detail.

The computational cost of matrix operations

When working with bigger matrices, the work required to do various operations becomes important. To add two $n \times n$ matrices we have n^2 pairs of numbers that we must add. For the product of a scalar and an $n \times n$ matrix, we must multiply n^2 pairs of numbers. (If the entries are big, the multiplications are harder, but we ignore this for now, we call each of 'add' or 'multiply' or 'add and multiply' a *basic operation.*) We implicilty assume that we are working with $n \times n$ matrices and say that matrix addition and scalar multiplication both take n^2 operations, or n^2 time. We usually ignore constants, and think of $2n^2 \approx n^2$.

The dot product of two vectors of length n takes n multiplications and n additions, so we say it takes n operations. Multiplying an matrix by a vector requires n dot products, so n^2 operations. Multiplying two $n \times n$ matrics requires n matrix-vector multiplications, so costs n^3 operations.

Note

We can multiply two matrices in n^3 operations. This means that if your slow slow computer can do one operation per second then it takes about $2^3 = 8$ seconds to multiply two 2×2 matrices, and in an hour, or 3600 seconds, the best it can do is multiply two 15×15 matrices, (because $\sqrt[3]{3600} \approx 15$).

If we could reduce this to n^2 operations, then as $\sqrt{3600} = 60$, your computer could mutliply two 60×60 matrices in an hour. These differences can be important in real situations.

What does Gaussian Elimination cost? Let's count this for solving an $n \times n$ system.

- In elimination, to clear below the first pivot we must clear $n-1$ rows, and each row add a scalar multple of a vector to another vector, so takes $n+1$ operations. Together takes $(n+1)(n-1) = n^2 - 1$, let's say n^2, operations to clear under the first pivot.
- Clearing under the second pivot takes $(n-1)^2$, under the third $(n-2)^2$ and all together, elimination takes
$$2^2 + \cdots + n^2 = \sum_{i=2}^{n} i^2 \approx n^3/3 \approx n^3$$
operations.
- Back substitution is a bit faster. Clearing over the nth pivot only takes one operation per row, so $n-1$ operations, clearling over the second pivot takes $n-2$ operations, etc, and altogether

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back substitution takes

$$\sum_{i=1}^{n-1} i \approx n^2.$$

- The whole process takes about $n^3 + n^2 \approx n^3$ operations. (Compared to elimination, back substitution is cheap!)

Note

Using Gauss-Jordan Elimination to find an inverse takes about twice as long, as you have to un-eliminate the augmented matrix I as you are eliminating A. But again, we usually ignore such constants.

The Transpose Matrix of A

For a matrix A, the *transpose* A^T is the matrix we get by switching rows and columns. For example:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \text{ and } \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 2 & 3 \end{bmatrix}^T = \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

It is not hard to see that

- $(AB)^T = B^T A^T$ and
- $(A^{-1})^T = (A^T)^{-1}$.

A matrix A is *symmetric* if $A^T = A$.

For any matrix A, AA^T is (square and) symmetric. Indeed,

$$(AA^T)^T = (A^T)^T A^T = AA^T.$$

Similarily $A^T A$ is symmetric.

Practice

Find an example to show that $A^T A = AA^T$ does not generally hold.

Problems from the text

1.6: 1, 2, 4, 6, 16, 21, 25, 27, 39, 49, 56 (9, 5, 10, 4, 13, 25, 28, 24, 38, 51, 60)

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We always use $\mathbf{x}$ for the column vector $\begin{bmatrix} x \\ y \end{bmatrix}$ or $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$, etc. of n variables.

1.5 Triangular Factors and Row Exchanges

We saw that by finding the inverse of A (in time n^3), we can then solve the matrix equation $Ax = b$ quickly (in time n^2) for several different vectors b. But we don't actually have to completely find A^{-1}. It is enough to eliminate, and skip the back substitution.

Doing elimination to $[A \mid I]$ we get to $[I \mid A^{-1}]$. But stopping before back-substitution, we get a matrix $[U \mid L^{-1}]$ where U is upper-triangular and L^{-1} is lower triangular.

This means that $L^{-1}A = U$, and applied to the system $A\mathbf{x} = \mathbf{b}$ gives

$$U\mathbf{x} = L^{-1}A\mathbf{x} = L^{-1}\mathbf{b} =: \mathbf{c}.$$

So we can solve for $\mathbf{x}$ by (i) letting $\mathbf{c} = L^{-1}\mathbf{b}$, and (ii) solving $U\mathbf{x} = \mathbf{c}$.

Example 1.12. We eliminate the augmented matrix $[A \mid I]$ below:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 8 & 0 & 1 & 0 \\ 2 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & -3 & -4 & -2 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 2 & -2 & 1 & 0 \\ 0 & 0 & 2 & -8 & 3 & 1 \end{array}\right] =: [U \mid L^{-1}]$$

It is important that we only use elementary row operations here, not row swaps, if we use row sways, then U may not be upper triangular.

Now to solve, say, $A\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ we compute

$$\mathbf{c} = L^{-1}\mathbf{b} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -8 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -3 \end{bmatrix}$$

and then solve

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & -3 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 0 & 11/2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -3/2 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1/2 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & -3/2 \end{array}\right]$$

to get $\mathbf{x} = (-1, 6, -3)/2$.

Computationally, this is not really a great savings: computing $L^{-1}\mathbf{b}$ is a matrix-vector multiplication, so takes n^2 operations (but actually only half of this as L^{-1} is triangular), and U is already eliminated, so solving $U\mathbf{x} = \mathbf{c}$ is just back-substitiution, so takes $n^2/2$ operations. This is about the same as computing $A^{-1}\mathbf{b}$. But there are some nice theoretical uses of L and U.

The LU-factorisation

Apart from the use we just saw, this partial elimination $[A \mid I] \to [U \mid L^{-1}]$ will have other uses. As $L^{-1}A = U$ we can write A as

$$A = LU$$

where L is lower-triangular and U is upper-triangular.

This LU-factorisation of a matrix A, if it exists, need not be unique:

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$$A = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ 4 & 2 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 2/3 & 8 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 2/3 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 3 & 6 \\ 0 & 0 & 2 \end{bmatrix}$$

This suggests an LDU-factorisation, in which D is diagonal, and L and U have only 1s on the diagonal. This is unique, if it exists, and is also used later.

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 4 & 2/3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

Practice

Show that if A is symmetric and has an LDU-factorisation, then it is of the form $A = LDL^T$.

Permutation Matrices

We saw that Gaussian Elimination can fail for some matrices A, we also saw that we might need to use 'row swaps', and we said that they are not 'elementary row operations'. The matrix that we multiply on the left to give a row swap is called a permutation matrix. A *permutation matrix* is an matrix that we can get from the identity by doing row swaps.

Practice

Which of the following matrices are permutation matrices?

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \end{bmatrix}$$

Practice

How many $n \times n$ permutation matrices are there?

When we are doing elimination, if we have to do a row swap, we can pretend that we did it before we started the elimination. (The other row operations might change, but we will still be able to do them.) So if we can eliminate A to I using row swaps, then we can eliminate PA to I without using row swaps, for some permutation matrix P.

Thus if A is invertable, then there are U, L and P such that

$$PA = LU.$$

Practice

Show, for a permutation matrix P, that $P^{-1} = P^T$.

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Problems from the text

1.5: 1, 2, 7, 10, 15, 18 (3, 1, 5, 6, 12, 11)

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2 Vector Spaces

2.1 Vector Spaces and Subspaces

Recall that we called an n-tuple $(r_1, r_2, \ldots, r_n)$ of elements of $\mathbb{R}^4$ a 'vector'. The reason we use the word vector is that the set $\mathbb{R}^4$, with scalar multiplication and componentwise addition, satisfies the following definition of a vector space.

Definition 2.1. A set X with addition '+' and scalar multiplication $\cdot : \mathbb{R} \times X \to X$ is a *(real) vector space* if it satisfies the following properties for all $x, y, z \in X$ and $q, r \in \mathbb{R}$.

a) $x + y = y + x$ for all $x, y \in X$

b) $x + (y + z) = (x + y) + z$ for all $x, y, z \in X$

c) There is an element $0 \in X$ such that $0 + x = x$ for all $x \in X$. This element is called zero.

d) For every $x \in X$ there is a element $-x \in X$ such that $x + (-x) = 0$.

e) $1 \cdot x = x$ for all $x \in X$.

f) $(qr)x = q(rx)$ for $x \in X$ and $q, r \in \mathbb{R}$.

g) $r(x + y) = rx + ry$ for all $x, y \in X$ and $r \in \mathbb{R}$.

h) $(q + r)x = qx + rx$ for all $x \in X$ and $q, r \in \mathbb{R}$.

Elements of a vector space X are called *vectors*.

Practice

Verify $\mathbb{R}^n$ is a vector space.

As we said, $\mathbb{R}^n$ is a vector space, and it is the one we talk about the most. But there are other real vector spaces:

Example 2.2. The following are vector spaces.

- $\emptyset$
- $\mathbb{R}^\infty$, the space of all infinite tuples: $(1, 2, 1, 1, 3, \ldots,)$. (Addition and scalar multiplication are again componentwise.)
- The space of 2×2 real matrices. (This is actually isomorphic as a space to $\mathbb{R}^4$. Matrix multiplication is not part of the space, it will define operations on the space.)
- The space of functions (like $f : x \mapsto 2x$, say) on a set $D \subset \mathbb{R}$. Addition is defined by $[f + g](x) = f(x) + g(x)$ and scalar multiplication by $[r \cdot f](x) = r \cdot f(x)$.

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Note

An *isomorphism* of vector spaces U and V is a bijective function $f : U \to V$ such that

- $f(a + b) = f(a) + f(b)$, and
- $f(ra) = rf(a)$.

If there is an isomorphism between two spaces, they are *isomorphic*.

Subspaces

A *subspace* of a vector space V is a subset U of V that is *closed under addition and scalar multiplication:*

- $x, y \in U \Rightarrow x + y \in U$, and
- $x \in U$, $r \in \mathbb{R} \Rightarrow rx \in U$.

A subspace of a vector space is itself a vector space.

Practice

In $\mathbb{R}^2$, is the subset $U = \{(1, 1), (1, -1)\}$ a subspace?

Example 2.3. Some subspaces of $\mathbb{R}^2$ are

- $\emptyset$, $\mathbb{R}^2$.
- $U_1 = \{(r, 0) \mid r \in \mathbb{R}\}$
- $U_2 = \{(0, r) \mid r \in \mathbb{R}\}$
- $U_3 = \{(r, 2r) \mid r \in \mathbb{R}\}$

Practice

Show that $U_3 = \{(r, 2r) \mid r \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.

These subspaces are sets, so we can graph them.

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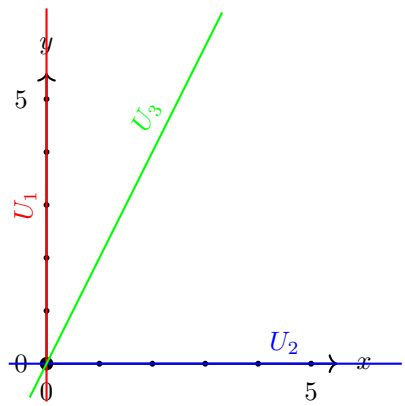

In general in $\mathbb{R}^2$ if a subspace contains a vector (a, b) then it must contain all scalar multiples $r(a, b)$ of (a, b) so must contain the line through $0(a, b) = (0, 0)$ and (a, b).

As soon as it also contains some (a', b') not on this line, what does it contain?

The only subspaces of $\mathbb{R}^2$ are $\mathbb{R}^0 = \{\mathbf{0}\}, \mathbb{R}^2$ and lines through $\mathbf{0} = (0, 0)$. A line $\{r(a, b) \mid r \in \mathbb{R}\}$ through $\mathbf{0}$ is isomorphic to $\mathbb{R}$ via the isomorphism

$$r \mapsto r(a, b).$$

So though $\mathbb{R}^1$ is not a subspace of $\mathbb{R}^2$ it is isomorphic to one. The subspaces of $\mathbb{R}^2$ are, **up to isomorphism**, $\mathbb{R}^0, \mathbb{R}^1$, or $\mathbb{R}^2$.

Practice

Up to isomorphism, what are the subspaces of $\mathbb{R}^n$?

2.1.1 The subspace generated by a set

We saw that if (a, b) is in a subspace U of $\mathbb{R}^2$ then the whole line $\{r(a, b) \mid r \in \mathbb{R}\}$ must be in U.

For a subset S of vectors in a vector space V, the *subspace generated by* S is the smallest subspace of V that contains S.

Practice

Show that if U and U' are two subspaces of V then so is their intesection $U \cap U'$. Use this to assert that for any subsets S of V, the subspace generated by S is uniquely defined.

Example 2.4. The subspace of $\mathbb{R}^4$ generated by $\{(0, 1, 2, 1)\}$ is the line

$$\{r(0, 1, 2, 3) \mid r \in \mathbb{R}\}.$$

The subspace of $\mathbb{R}^3$ generated by $\{(0, 1, 2), (1, 1, 0)\}$ is the plane

$$\{a(0, 1, 2) + b(1, 1, 0) \mid a, b \in \mathbb{R}\};$$

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it is isomorphic to $\mathbb{R}^2$. (You can show that $(a, b) \mapsto a(0, 1, 2) + b(1, 1, 0)$ is an isomorphism.)

We will see that every subspace of $\mathbb{R}^n$ is isomorphic to $\mathbb{R}^m$ for some non-negative $m \leq n$. This m is the *dimension* of the subspace.

The Column Space of a Matrix

As the columns of an $m \times n$ matrix M are vectors in $\mathbb{R}^m$, they generate a subspace of $\mathbb{R}^m$. This is called the the *column space* $C(M)$ of M.

Example 2.5. The column space of

$$\begin{bmatrix} 0 & 4 \\ 1 & 2 \\ 5 & 3 \end{bmatrix} \text{ is } \left\{ u \begin{bmatrix} 0 \\ 1 \\ 5 \end{bmatrix} + v \begin{bmatrix} 4 \\ 2 \\ 3 \end{bmatrix} \mid u, v \in \mathbb{R} \right\}.$$

The matrix equation

$$\begin{bmatrix} 0 & 4 \\ 1 & 2 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 7 \\ 2 \\ 1 \end{bmatrix}$$

has a solution if and only if $\left[\begin{smallmatrix} 7 \\ 2 \\ 1 \end{smallmatrix}\right]$ is in this space.

Observe that the column space of M has a useful interpretation as the set of vectors $\mathbf{b}$ such that the equation $M\mathbf{x} = \mathbf{b}$ has a solution. From this we conclude that if $M\mathbf{x} = \mathbf{b}$ and $M\mathbf{x} = \mathbf{b}'$ both have solutions, then so does $M\mathbf{x} = \mathbf{b} + \mathbf{b}'$.

Practice

Show directly, without using the observation above, that the set S of vectors $\mathbf{b}$ such that $M\mathbf{x} = \mathbf{b}$ has a solution, is a subspace of $\mathbb{R}^m$.

The Nullspace of a Matrix

As the solutions of

$$\begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

are vectors in $\mathbb{R}^2$, the solutions of $M\mathbf{x} = \mathbf{0}$ for an $m \times n$ matrix are vectors in $\mathbb{R}^n$. Observing that if

We let $\mathbf{0}$ denote the appropriate vector or matrix of all '0's.

$$M\mathbf{x} = \mathbf{0} \text{ and } M\mathbf{x}' = \mathbf{0}$$

then

$$M\mathbf{x} + \mathbf{x}' = M\mathbf{x} + M\mathbf{x}' = \mathbf{0} + \mathbf{0} = \mathbf{0},$$

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and

$$M(r\mathbf{x}) = (M\mathbf{x})r = \mathbf{0}r = \mathbf{0},$$

we see that the set of solutions to $M\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbb{R}^n$. This space is called the *nullspace* $N(M)$ of M.

Think of the nullspace of M as the set of vectors that M nullifies, or 'takes to zero'.

Problems from the text

2.1: 2, 3, 5, 6, 8, 13, 17, 25, 30 (2, 4, 3, 5, 8, 12, 14, 26, 29)

2.2 Solving $M\mathbf{x} = \mathbf{0}$ and $M\mathbf{x} = \mathbf{b}$

When a matrix M has an inverse there is a unique solution $M^{-1}\mathbf{b}$ to $M\mathbf{x} = \mathbf{b}$, and we know how to solve this. But M might not be square, and if it is, it might be singular (non-invertible). The equation $M\mathbf{x} = \mathbf{b}$ may have no solutions, or many solutions. We want to find all of them. As expected, we do this using Gaussian elimination.

Echelon Form of a matrix

Consider the matrix $M = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix}$. We reduce it using Gaussian Elimination:

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix} \rightsquigarrow \underbrace{\begin{bmatrix} 1 & 3 & 3 & 2 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Echelon Form}} \rightsquigarrow \underbrace{\begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}}_{\text{Row Reduced Echelon Form}}$$

The symbol $\rightsquigarrow$ hides a lot of row operations.

Echelon Form is as 'close' as we can get to triangular: we try to make the system triangular but we may get unavoidable 0's on the diagonal. The Row Reduced Echelon Form (RREF) is a 'close' as we can get to diagonal. The first non-zero entry in a row is a *pivot*, the column it is in is a *pivot column*. The other columns are *non-pivot* or *free* columns. To find the set of solutions to $M\mathbf{x} = \mathbf{b}$ using the RREF of M the pivot columns and free columns treated differently.

Solving $M\mathbf{x} = \mathbf{0}$

To solve $M\mathbf{x} = \mathbf{0}$ we have reduced to

$$\begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}.$$

The variables in $\begin{bmatrix} w & x & y & z \end{bmatrix}$ corresponding to pivot and free columns are called *dependent* and *independent (or free)* respectively. So w and y are dependent and x and z are free.

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The free variables can take any value and still yield a solution, so we leave them as variables: $x = x$ and $z = z$. Our solution will look like

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ \\ 1 \end{bmatrix}.$$

With x and z fixed, the dependent variables are determined. The second row contains the pivot corresponding to y so allows us to solve for y:

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}.$$

The first row allows us to solve for w:

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = x \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

This is a solution for any choice of $x, z \in \mathbb{R}$. So we have found the nullspace of M.

The number of pivots in M is an important number. The *rank* $r \leq n$ of an $m \times n$ matrix M is the number of pivots when it has been reduced to Echelon Form. The number, $n - r$, of free variables is the *nullity* of M. This is the number of columns in our solution to $M\mathbf{x} = \mathbf{0}$. It will be the dimension of our nullspace.

Solving $M\mathbf{x} = \mathbf{b}$

Unlike the set of solutions of $M\mathbf{x} = \mathbf{0}$, the set of solutions of $M\mathbf{x} = \mathbf{b}$ need not be a space. However, if we find one solution $\mathbf{x_0}$ to $M\mathbf{x} = \mathbf{b}$ then $\mathbf{x_0} + \mathbf{z}$ is another for any solution $\mathbf{z}$ of $M\mathbf{x} = \mathbf{0}$. Indeed

$$M(\mathbf{x_0} + \mathbf{z}) = M\mathbf{x_0} + M\mathbf{z} = \mathbf{b} + \mathbf{0} = \mathbf{b}.$$

The set of solutions of $M\mathbf{x} = \mathbf{b}$ is called its *complete solution*. We will see that it is

$$S = \{\mathbf{x_0} + z \mid z \in \text{nullspace(M)}\}$$

for any one *particular solution* $\mathbf{x_0}$ of $M\mathbf{x} = \mathbf{b}$.

Example 2.6. We find the complete solution of

$$\begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix} \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix} =: \mathbf{b}$$

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by eliminating the augmented matrix to RREF:

$$\left[\begin{array}{cccc|c} 1 & 3 & 3 & 2 & 1 \\ 2 & 6 & 9 & 7 & 5 \\ -1 & -3 & 3 & 4 & 5 \end{array}\right] \to \left[\begin{array}{cccc|c} 1 & 3 & 0 & -1 & -2 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right].$$

Again we solve: $z = z$, $y = 1 - z$, $x = x$, and $w = -2 - 3x + z$. So we have complete solution

$$\begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix} = \underbrace{\begin{bmatrix} -2 \\ 0 \\ 1 \\ 0 \end{bmatrix}}_{\text{particular soln}} + x \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + z \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix}.$$

Problems from the text

2.2: 4, 5, 7, 10, 18, 29, 38, 41, 53, 65 (5, 2, 6, 12, 20, 30, 38, 39, 54, 60)

2.3 Linear Independence

The subspace of $\mathbb{R}^4$ generated by $\{(1,1,1,0),(1,1,0,0),(-1,-1,1,0)\}$ can also be generated by the set $\{(0,0,1,0),(1,1,0,0)\}$. This smaller generating set is more desirable than the big one for obvious reasons. How do we find the smallest generating set of a space?

Definition 2.7. A *linear combination* of vectors $v_1, \dots v_k$ is a sum of the vectors, each multiplied by some scalar:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k.$$

A set $\{v_1, \dots, v_k\}$ of vectors is *linearly independent* if no one of them can be written as a linear combination of the others.

Practice

Show that a set $\{v_1, \dots, v_k\}$ of vectors is linearly independent if and only if

$$(c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0) \Rightarrow (c_1 = c_2 = \cdots = c_k = 0).$$

Example 2.8. Is $\{(1,1,1),(2,3,4),(0,1,2)\}$ independent?

No! $(2,3,4) - (1,1,1) = (0,1,2)$.

Is $\{(1,1,1),(2,3,4)\}$ independent?

Yes! If $a(1,1,1) + b(2,3,4) = 0$ then

$$\begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

which we can quickly show yields $(a,b) = (0,0)$.

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We can generalise this to the following assertion:

Fact 2.9. *A set of vectors is linearly independent if and only if the matrix M of which they are columns yields a unique solution to $M\mathbf{x} = \mathbf{0}$.*

Example 2.10. Are the vectors $(1, 1, 2), (3, 2, 2)$, and $(2, -1, 4)$ linearly independent?

Well, eliminating, we get

$$\begin{bmatrix} 1 & 3 & 2 \\ 1 & 2 & -1 \\ 2 & 2 & 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix},$$

so yes.

Example 2.11. What if we include $(7, 0, -8)$ also?

No! A 3×4 matrix can have at most 3 pivots. It will have a free variable, so $M\mathbf{x} = \mathbf{0}$ will not have just one solution.

Example 2.12. OK. Then we should be able to express $\mathbf{b} = (7, 0, -8)$ as a linear combination of the other three vectors. How do we do this?

One method: We could solve $M\mathbf{x} = \mathbf{b}$.

$$\left[\begin{array}{ccc|c} 1 & 3 & 2 & 7 \\ 1 & 2 & -1 & 0 \\ 2 & 2 & 4 & -8 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & -10.5 \\ 0 & 1 & 0 & 5.5 \\ 0 & 0 & 1 & .5 \end{array}\right]$$

So $\left[\begin{smallmatrix} 7 \\ 0 \\ -8 \end{smallmatrix}\right] = -10.5\left[\begin{smallmatrix} 1 \\ 1 \\ 2 \end{smallmatrix}\right] + 5.5\left[\begin{smallmatrix} 3 \\ 2 \\ 2 \end{smallmatrix}\right] + .5\left[\begin{smallmatrix} 2 \\ -1 \\ 4 \end{smallmatrix}\right]$.

Another method: Put the vectors as rows and eliminate the following augmented matrix.

$$\left[\begin{array}{ccc|cccc} 1 & 1 & 2 & 1 & 0 & 0 & 0 \\ 3 & 2 & 2 & 0 & 1 & 0 & 0 \\ 2 & -1 & 4 & 0 & 0 & 1 & 0 \\ 7 & 0 & -8 & 0 & 0 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|cccc} 1 & 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & 4 & 3 & -1 & 0 & 0 \\ 0 & 0 & 12 & 7 & -3 & 1 & 0 \\ 0 & 0 & 0 & 10.5 & -5.5 & -.5 & 1 \end{array}\right]$$

Hmm.

Practice

How does this give us the answer?

Note

A nice thing to notice about this second method is that all the reduced rows are in the space generated by the original rows. They seem to give us a nicer generating set of the same space.

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Definition 2.13. A set of vectors S *span* a vector space V if V is the space they generate. The set S is a *basis* for V if it spans V and is linearly independent.

Example 2.14. The set $\{e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)\}$ is a basis for $\mathbb{R}^3$. Indeed, it is called the *standard basis*.

Practice

Find another basis for $\mathbb{R}^3$.

Practice

Find one containing $(10, 10, -2)$.

Fact 2.15. *Given a basis B of a vector space V, a vector v in V has a unique expression as a linear combination of vectors in B.*

Practice

Why is this true?

'Basis' is singular, 'bases' is plural.

Though a space can have different bases, they must have the same size.

Theorem 2.16. *Different bases of the same vector space have the same number of vectors.*

Proof. Assume that $\{b_1, \dots b_m\}$ and $\{a_1, \dots, a_n\}$ are two different bases of a vector space $V \subset \mathbb{R}^d$. As every a_i is a linear combination of the b_i there is an $m \times n$ matrix M such that

$$\underbrace{[b_1|b_2|\dots|b_m]}_{B} M = \underbrace{[a_1|a_2|\dots|a_n]}_{A}.$$

If $n > m$ then $M\mathbf{x} = \mathbf{0}$ has a non-zero solution $\mathbf{x_0}$ and so

$$A\mathbf{x_0} = BM\mathbf{x_0} = B\mathbf{0} = \mathbf{0}$$

is a linear combination of the a_i that equals 0. This is impossible as $\{a_1, \dots, a_n\}$ is a basis, and so $n \leq m$. Similarily we can show that $m \leq n$. $\square$

Definition 2.17. The number of vectors in a basis of a vector space is its *dimension*.

Problems from the text

2.3: 3, 4, 7, 9, 10, 18, 21, 30, 34, 36 (5, 1, 2, 10, 8, 18, 26, 33, 30, 29)

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2.4 The Four Fundamental Subspaces

For a matrix M we have seen the column space $C(M)$ and the nullspace $N(M)$. We can find bases for these spaces by elimination. Lets eliminate some M to some upper triangular U:

$$M = \begin{bmatrix} 1 & 3 & 3 & 2 \\ 2 & 6 & 9 & 7 \\ -1 & -3 & 3 & 4 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 3 & 0 & -1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} =: U.$$

Basis of the nullspace

As $M\mathbf{x} = \mathbf{0} \iff U\mathbf{x} = \mathbf{0}$, M and U have the same nullspace

$$N(M) = N(U) = \left\{ z \begin{bmatrix} 1 \\ 0 \\ -1 \\ 1 \end{bmatrix} + x \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix} \mid x, z \in \mathbb{R} \right\}$$

having dimension $n - r$, and basis $B_{N(M)} = \{(1, 0, -1, 1), (-3, 1, 0, 0)\}$.

Basis of the column space

The column spaces $C(M)$ and $C(U)$ are different. (This is easy to see for, say, the matrix $M = \left[\begin{smallmatrix} 1 \\ 1 \\ 1 \end{smallmatrix}\right]$ which would eliminate to $C = \left[\begin{smallmatrix} 1 \\ 0 \\ 0 \end{smallmatrix}\right]$.) The column space $C(U)$ of U is easy to read off: it is generated by the pivot columns, so has dimension $r = 2$ and basis $B_{C(U)} = \{(1, 0, 0), (0, 1, 0)\}$.

Now because $M\mathbf{x} = \mathbf{0} \iff U\mathbf{x} = \mathbf{0}$, columns of M are independent if and only if the same columns of C are independent. As the pivot columns of U were the first and third, and these gave a basis of $C(U)$ the first and third columns of M are a basis of $C(M)$. It has dimension $r = 2$ and basis $B_{C(A)} = \{(1, 2, -1), (3, 9, 3)\}$.

The row space of M

The *row space* of an $m \times n$ matrix M is the subspace of $\mathbb{R}^n$ generated by the rows of M. These are the columns of M^T so we denote it by $C(M^T)$.

Clearly M and U have the same row space; and it is generated by the pivot (non-zero) rows of U. So $B_{C(M^T)} = \{(1, 3, 0, -1), (0, 0, 1, 1)\}$. It has dimension r.

The left nullspace of M

The *left nullspace* of M is the space of row vectors $\mathbf{x} \in \mathbb{R}^m$ such that $\mathbf{x}M = \mathbf{0}$. This is the nullspace of M^T so we denote it $N(M^T)$.

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As $N(M^T)$ is the nullspace of M^T if has dimension $m-r$. We can again get the basis by elimination. Computing

$$\left[\begin{array}{cccc|ccc} 1 & 3 & 3 & 2 & 1 & 0 & 0 \\ 2 & 6 & 9 & 7 & 0 & 1 & 0 \\ -1 & -3 & 3 & 4 & 0 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cccc|ccc} 1 & 3 & 3 & 2 & 1 & 0 & 0 \\ 0 & 0 & 3 & 3 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 5 & -2 & 1 \end{array}\right] =: [U \mid L],$$

We see that $\begin{bmatrix} 5 & -2 & 1 \end{bmatrix} A = \mathbf{0}$, so this row is in the left nullspace. More generally, rows of L corresponding to the non-pivot rows of M are in the nullspace, and there are exactly $m-r$ of them, so they are a basis for $N(M^T)$. So $B_{N(M^T)} = \{(5,-2,1)\}$.

Left and right inverses

A *left inverse* of an $m \times n$ matrix M is an $n \times m$ matrix L such that

$$LM = I_{n\times n}.$$

A *right inverse* is an $n \times m$ matrix R such that

$$MR = I_{m\times m}.$$

If such an L exists, then the rows of $I_{n\times n}$ are in the $N(M^T) \leq R^n$. As they are the standard basis of R^n we have then that $N(M^T) = R^n$ and so

$$n = r \leq m.$$

In particular, the matrix is *tall*: $n \leq m$. Similarily, we can show that if such an R exists then

$$m = r \leq n.$$

In particular, the matrix is *wide*: $n \geq m$. If both exist then $m = n$ and (as we saw in Problem #21 of Section 2.2), $L = R$ is the inverse of M.

Observe that if M has a left inverse L then L^T is the right inverse of A^T:

$$(A^T L^T) = (LA)^T = I^T = I.$$

Finding the left inverse

We saw that the left inverse of $m \times n$ matrix M exists only if $n = r \leq m$. Indeed, this is 'if and only if'. The following example shows how one can find a left inverse for a tall matrix M: one where $n \leq m$. Observe how if this fails, we will have that $r < n$.

Example 2.18. By Gauss Jordan, reduce $[M \mid I]$:

$$\left[\begin{array}{ccc|cccc} 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 3 & 2 & 1 & 0 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 & 0 & 1 & 0 \\ 3 & 3 & 2 & 0 & 0 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{ccc|cccc} 1 & 0 & 0 & -4 & 2 & 1 & 0 \\ 0 & 1 & 0 & 7 & -3 & -2 & 0 \\ 0 & 0 & 1 & -2 & 1 & 1 & 0 \\ 0 & 0 & 0 & -5 & 1 & 1 & 1 \end{array}\right] =: [U \mid B]$$

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As

$$\begin{bmatrix} -4 & 2 & 1 & 0 \\ 7 & -3 & -2 & 0 \\ -2 & 1 & 1 & 0 \\ -5 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 3 & 2 & 1 \\ -1 & 0 & 2 \\ 3 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix},$$

the first three rows of B make a left inverse L of M. So $L = \left[\begin{smallmatrix} -4 & 2 & 1 & 0 \\ 7 & -3 & -2 & 0 \\ -2 & 1 & 1 & 0 \end{smallmatrix}\right]$.

Notice also that $(-5, 1, 1, 1)$ is in $N(M^T)$. So adding it to any row of L doesn't change the fact that it is a left inverse of M. So there can be may left inverses of M.

Finding the right inverse

The right inverse exists only if the matrix is wide. The following example shows that it exists if $m = r$.

Example 2.19. Reduce M by elimination

$$\left[\begin{array}{cccc|ccc} 1 & 1 & 2 & 3 & 1 & 0 & 0 \\ 1 & 2 & 0 & 3 & 0 & 1 & 0 \\ 1 & 3 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cccc|ccc} 1 & 0 & 0 & 7 & -2 & 7 & -4 \\ 0 & 1 & 0 & -2 & 1 & -3 & 2 \\ 0 & 0 & 1 & -1 & 1 & -2 & 1 \end{array}\right] =: [U \mid B].$$

As $m = r$ some choice of m columns of U contains $I_{m\times m}$. Choosing the corresponding columns M' of M we get

$$\begin{bmatrix} -2 & 7 & -4 \\ 1 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix}}_{M'} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

So $B = M'^{-1}$ is the inverse of the square matrix M'. So it is also the right inverse of M':

$$\begin{bmatrix} 1 & 1 & 2 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} -2 & 7 & -4 \\ 1 & -3 & 2 \\ 1 & -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We want a right inverse of M though. So when we add the column back to M' we have to add a row of 0s back to B to get R:

$$\begin{bmatrix} 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 3 \\ 1 & 3 & 1 & 2 \end{bmatrix} \begin{bmatrix} -2 & 7 & -4 \\ 1 & -3 & 2 \\ 1 & -2 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Here we have found a right inverse R of the wide matrix M, and the only thing we needed was that there were m pivots. This works whenever $m = r$. Observe also that as M is wide, it has non-trivial nullspace. Adding vectors from the $N(M)$ to the columns of R does not change the product, so gives other right inverses.

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Problems from the text

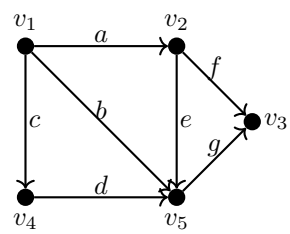
2.4: 2, 4, 6, 9, 11, 15, 23, 28, 30, 31, 33 (3, 5, 12, 14, 17, 16, 27, 29, 33, 34, 30)

2.5 Graphs applications

A graph consists of a set of *vertices* (or *nodes*) $V = \{v_1, \ldots, v_n\}$, and a set E of two element subsets of V, called *edges*.

For us though, when we say 'graph' we mean *'oriented graph'*: the edges are ordered pairs rather than sets. So we usually call them *arcs* instead of *edges*. We cannot have both the edge (u, v) and the edge (v, u).

Example 2.20. We can draw a graph G as follows.



and can alternately represent it by its incidence matrix M_G:

$$\begin{array}{cc} & \text{Nodes} \\ & \begin{array}{ccccc} v_1 & v_2 & v_3 & v_4 & v_5 \end{array} \\ \text{Arcs} \begin{array}{c} a \\ b \\ c \\ d \\ e \\ f \\ g \end{array} & \begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix} \end{array}$$

We interpret the spaces associated to the incidence matrix M_g of a graph G.

Nullspace of M_G

Observe that the constant vector of ones is in the nullspace $N(M_G)$. Indeed, each row contains a 1 and -1 so the dot product with a vector $\vec{1}$ of ones is $1 - 1 = 0$. What else is in the nullspace? Well for any vector $x = (x_1, \ldots, x_n)$ in the nullspace we have that $x - (x_1 - 1)\vec{1}$ is also in the nullspace, so we may assume that $x_1 = 1$. Now, if there is an arc a from v_1 to v_i, then we get that $x_i = 1$ form the fact that the row of M corresponding to a has a -1 in the v_1 column and 1 in the v_i

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column. In the graph shown above we can then repeat this argument, and show that $x_i = 1$ for all i. So M_G has nullity 1. But this is not true of all graphs. A *path* in G is a sequence of distinct vertices $v_1, v_2, \ldots, v_\ell$ such that for all i either (v_i, v_{i+1}) or (v_{i+1}, v_i) is an arc in G. Where $u = v_1$ and $v = v_\ell$ it is a path *between* u and v. A *component* in a graph G is a maximal subset C of the vertices such that for any two vertices u and v in C there is a path in G between u and v. If G has only one component, it is *connected*.

Practice

Show that for any connected graph G, M_G has nullity 1. Show that in general the nullity of M_G is the number of components of G.

Rowspace and Left Nullspace

A *cycle* in a graph is a path $v_1, \ldots, v_\ell$ **except** that $v_1 = v_\ell$. We often describe a cycle just by its arcs. If (v_i, v_{i+1}) is an arc it is a *forward* arc in the cycle, and if (v_{i+1}, v_i) is, it is a *backward arc*. The characteristic vector χ_C of a cycle C is the vector $(x_1, \ldots, x_m)$ such that $x_i = 1$ if e_i is a forward arc in C, $x_i = -1$ if e_i is a backward are in C, and otherwise, $x_i = 0$.

One will notice the following.

Fact 2.21. *For a cycle C of G*

$$\chi_C M_G = 0.$$

Example 2.22. The cycle $a + e - b$ that goes from v_1 to v_2 to v_5 and back to v_1 corresponds to the product

$$\begin{bmatrix} 1 & -1 & 0 & 0 & 1 & 0 & 0 \end{bmatrix} M_G = \begin{matrix} 1 \begin{bmatrix} 1 & -1 & 0 & 0 & 0 \end{bmatrix} + \\ -1 \begin{bmatrix} 0 & -1 & 0 & 0 & 1 \end{bmatrix} + \\ 1 \begin{bmatrix} -1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix} = \mathbf{0}$$

Fact 2.21 tells us that **cycles of G are vectors in the left nullspace of M_G**. As the properties of a space are useful, we often generalise the definition of cycles as any set of arcs corresponding to a vector in the left nullspace of M_G. Thus we can add cycles, and so get a *cycle space of G*.

Example 2.23. Where C_1 is the cycle $a + e - b$ and C_2 is $b - d - c$ the sum $C_1 + C_2 = a + e - d - c$.

Practice

Exemplify this sum of cycles with a picture.

Practice

Show that under this definition, cycles need not be connected.

We also see from Fact 2.21 that the rows of M_G corresponding to the arcs of a cycle of G are dependent. One can prove the following.

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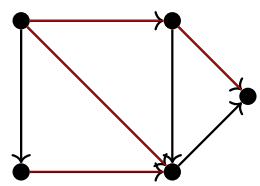
Fact 2.24. *The arcs corresponding to any dependent set of rows of M_G contain a cycle in G. If the co-efficients in the dependence are all ± 1, then M_G is a (union of disjoint) cycles.*

The dimension r of the rowspace is the size of the biggest subgraph with no cycles. In a connected graph this is a spanning tree,



which has $r = n - 1$ arcs. In general $r = n - c(G)$ where $c(G)$ is the number of components of G. It follows that the dimension of the column space is $n - r = c(G)$, as we observed above.

Practice

Show that the dimension of the cycles space of G is $m - n + c(G)$?

An application: Euler's Formula

We (non-rigorously) prove a (relatively easy) classical result of topology / graph theory.

A graph is *planar* if it can be drawn on the plane without any crossing edges. Take a plane drawing of a planar graph. Cutting the plane up by cutting along the edges you partition the plane into the *faces* of the the graph. The edges separating a face from the rest of the plane are a cycle, and it is not hard to see that these 'face cycles' generate the cycle space of G (the left nullspace of M_G). (Generally in graph theory or topology we count the outer face as a face, but for today we do not!)

Thus the the number of faces is exactly the dimension of the cycle space, and so we have

$$f = m - n + c(G)$$

where f denotes the number of faces for the plane graph G. This is Euler's Formula.

The column space

Recall that the column space of a matrix M is the set of $\mathbf{b}$ such that $M\mathbf{x} = \mathbf{b}$ has a solution. What does this mean when M is the adjacency matrix M_G of a graph G?

With our previously pictured G, let $\mathbf{b} = (2, 1, 1, 0, -1, 2, 3)$ and assume that $\mathbf{x} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{bmatrix}^T$ is a solution to $M_G\mathbf{x} = \mathbf{b}$. The solution assigns a value x_i to each node v_i, and $\mathbf{b}$ assigns a value to each arc.

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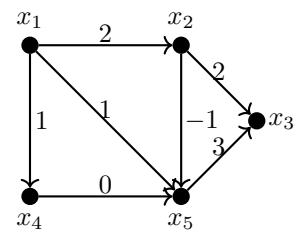
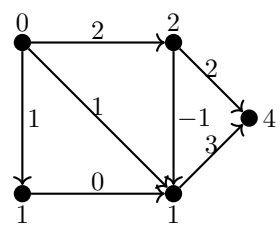


As $(1, 1, 1 \ldots, 1)$ is in the nullspace, if there is a solution, then there is one with $x_1 = 0$, so lets assume that $x_1 = 0$.

The first row of $M_G\mathbf{x} = \mathbf{b}$, that is, of

$$\begin{bmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 0 & 1 \\ 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 0 \\ -1 \\ 2 \\ 3 \end{bmatrix}$$

says that $-x_1 + x_2 = 2$. Thus the 2 corresponding on the arc $v_1 \to v_2$ means that $x_2 = x_1 + 2$. So a solution looks like this:



Were were lucky that this all worked out. When we follow the arcs around a circle, adding values to the vertices according to the values on the edges, things might not work out. We need that the sum of the arc values around any cycle (paying attention to orientation) is 0.

Fact 2.25. *$M_G\mathbf{x} = \mathbf{b}$ has a solution if and only if the sum of the values in $\mathbf{b}$ corresponding to any cycle c of G, add up to 0.*

Viewing the values x_i on each vertex as a voltage, and the values on the arcs as voltage drops (or as current when the wire between any two vertices is assumed to have resistance 1) then 'there is a solution to $M_G\mathbf{x} = \mathbf{b}$ if and only if the current around each closed walk adds up to 0.' You might recognise this as Kirchoff's law of conservation of energy.

Problems from the text

2.5: 2 (1)

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2.6 Linear Transformations

A function $T : V \to W$ of vector spaces is a *linear transformation* if it satisfies the following for all $a, b \in \mathbb{R}$ and $\mathbf{x}, \mathbf{y} \in V$:

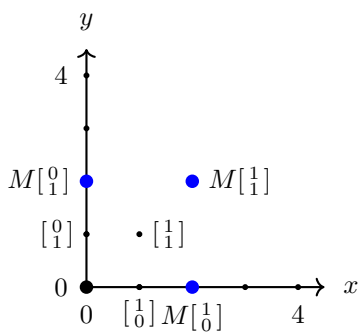
$$T(a\mathbf{x} + b\mathbf{y}) = aT(\mathbf{x}) + bT(\mathbf{y}).$$

Observe that matrix multiplication satisfies this:

$$M(a\mathbf{x} + b\mathbf{y}) = aM\mathbf{x} + bM\mathbf{y},$$

so multiplication (on the left) by a matrix M is a linear transformation.

Example 2.26. The matrix $\left[\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right]$ takes the point $(0, 1)$ to the point $\left[\begin{smallmatrix} 2 & 0 \\ 0 & 2 \end{smallmatrix}\right]\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right]$. It takes the point $(1, 1)$ to $(2, 2)$ and the point $(1, 0)$ to $(2, 0)$.



It seems to be 'blowing $\mathbb{R}^2$ up by a factor of 2', or 'zooming in on $\mathbb{R}^2$.

Practice

What does multiplication by 1) $\left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ 2) $\left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ 3) $\left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ do to these same points? Describe what multiplication by these matrices seems to be doing to $\mathbb{R}^2$ as a whole.

Practice

Which of the following are linear transformations of R^2?

i. Reflection in the x-axis.

ii. Mapping everything to 0.

iii. Shifting left 1.

For those that are, are there matrices that give this transformation?

A good way to find the matrix that gives a transformation, is to check first what the transformation does to a basis of the vector space.

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Example 2.27. Consider the transformation T of $\mathbb{R}^2$ in which every vector is reflected in the x-axis. On the standard basis we have $T((0,1)) = (0,-1)$ and $T((1,0)) = (1,0)$, so T is represented by the matrix M such that

$$M\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}0\\-1\end{bmatrix} \text{ and } M\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix}.$$

So $M = \left[\begin{smallmatrix}1 & 0\\0 & -1\end{smallmatrix}\right]$.

Rotation by $\pi/2$ degree is represented by the matrix $\left[\begin{smallmatrix}0 & -1\\1 & 0\end{smallmatrix}\right]$; what about rotation by an angle θ?

Practice

Represent the transformation R_θ which rotates everything in $\mathbb{R}^2$ by θ radians, by a matrix.

Now that you have finished the practice question (Do it if you haven't!) to rotate by 2θ we could multiply by R_θ twice, or by $(R_\theta)^2$ once. So rotation by 2θ is (writing c for $\cos\theta$ and s for $\sin\theta$.

$$\begin{bmatrix}\cos 2\theta & -\sin 2\theta\\ \sin 2\theta & \cos 2\theta\end{bmatrix} = (R_\theta)^2 = \begin{bmatrix}c & -s\\ s & c\end{bmatrix}\begin{bmatrix}c & -s\\ s & c\end{bmatrix} = \begin{bmatrix}c^2-s^2 & -2cs\\ 2cs & c^2-s^2\end{bmatrix}.$$

This gives us our double angle formulae:

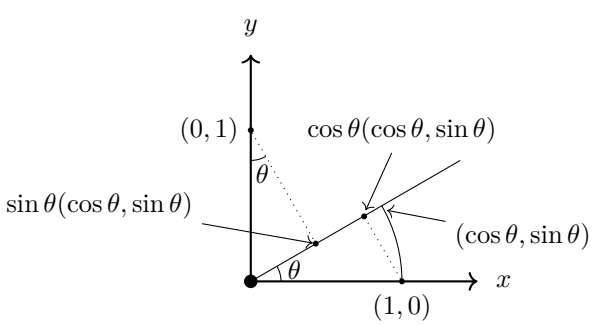
- $\cos 2\theta = \cos^2\theta - \sin^2\theta$
- $\sin 2\theta = 2\cos\theta\sin\theta$

Practice

Prove the angle sum identities for $\sin(\phi+\theta)$ and $\cos(\phi+\theta)$ in the same way.

Let's do some harder transformations.

Example 2.28. How do we represent the projection onto the line L_θ that makes a positive angle of θ with the x-axis? Well, we could figure out where the vector $(1,0)$ projects:



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And read off that this is represented by the matrix

$$\begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ cos\theta\sin\theta & \sin^2\theta \end{bmatrix}.$$

Or we get it another way, being clever instead of having to think. It is easy to find the matrix P_x that projects onto the x-axis: $P_x = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 0 \end{smallmatrix}\right]$. To project onto the line L_θ we can rotate a vector by $-\theta$, projecting onto the x-axis, and then rotating back by θ. So projection onto L_θ is

$$\begin{aligned} R_\theta P_x R_{-\theta} &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \cos -\theta & -\sin -\theta \\ \sin -\theta & \cos -\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos -\theta & -\sin -\theta \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} \cos^2\theta & \cos\theta\sin\theta \\ \cos\theta\sin\theta & \sin^2\theta \end{bmatrix} \end{aligned}$$

Multiplying by R_θ and R_θ^{-1} is called a change of basis. This is a useful technique that we well use later.

For the homework you will need to know that the *kernal* of a transformation T is the set of vectors v such that $T(v) = 0$. So it is the nullspace of the matrix M that represents the transformation T.

Problems from the text

2.6: 5, 9, 12, 16, 20, 23, 24, 26, 28, 29, 49 (6, 1, 3, 23, 22, 19, 17, 29, 26, 27, 50

3 Orthogonality

3.1 Orthogonal Vectors and Subspaces

We saw the standard basis for R^n:

$$\{\underbrace{(1,0,0,\ldots,0)}_{e_1}, \underbrace{(0,1,0,\ldots,0)}_{e_2}, \ldots, \underbrace{(0,0,0,\ldots,1)}_{e_n}\}.$$

It is a really nice basis to work with. When we are looking for a basis of some subspace U of R^n, we can't always get such a nice basis. What's nice about this basis? All the basis vectors have length (distance from the origin) 1, and they are all at right angles with each other. We try to get close to this for U.

Definition 3.1. Two non-zero vectors u and v in $\mathbb{R}^n$ are *orthogonal*, written $u \perp v$, if the (smallest) angle between them is $\pi/2$ (i.e., $90°$). It is useful to assume that the zero vector is orthogonal to all

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vectors. A vector is *unit* if it has length 1. A basis is *normal* if it consists of unit vectors, *orthogonal* if it consists of vectors are pairwise orthogonal, and *orthanormal* if it is normal and orthogonal. (Some books use othanogonal to mean orthanormal.)

The goal in this chapter is to take a basis, and find an equivalent orthanormal basis. We investigate many nice properties of orthogonality along the way.

Length of a vector

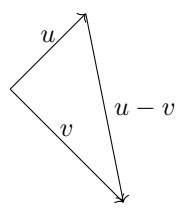
As the length of (x, y) in $\mathbb{R}^2$ is $\sqrt{x^2 + y^2}$ by pythagoras, the length of (x, y, z) in $\mathbb{R}^3$ is $\sqrt{x^2 + y^2 + z^2}$. The length of $v = (v_1, v_2, \ldots, v_n)$ is

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}.$$

So $\|v\| = \begin{bmatrix} v_1 & v_2 & \cdots & v_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = v^T v$, (viewing vectors as columns.) To get a basis of unit vectors from a basis, we can just replace each vector v by $v/\|v\|$. This is called *normalising* the vector space.

Orthogonal vectors

Now, vectors u and v are orthogonal



if and only if $\|u\|^2 + \|v\|^2 = \|u - v\|^2$. But

$$\begin{aligned} \|u - v\|^2 &= (u_1 - v_1)^2 + (u_2 - v_2)^2 + \cdots + (u_n - v_n)^2 \\ &= (u_1^2 - 2u_1v_1 + v_1^2) + (u_2^2 - 2u_2v_2 + v_2^2) + \ldots \\ &= (u_1^2 + \cdots + u_n^2) + (v_1^2 + \ldots, v_n^2) - 2(u_1v_1 + \cdots + u_nv_n) \\ &= \|u\|^2 + \|v\|^2 - 2u^T v \end{aligned}$$

So $u \perp v$ if and only if $u^T v = 0$.

Example 3.2.

- $(0, 0, 1)(0, 1, 0)^T = 0 + 0 + 0 = 0$: orthogonal.

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- $(1,0)(1,1)^T = 1 + 0 = 1$: not orthogonal.
- $(1,1)(1,-1)^T = 1 - 1 = 0$: orthogonal.

Note

Hey! is it $u^T v$ or uv^T? Sorry, yeah. When I write it as v we think of it as a column vector, so we write $u^T v$. But when I'm writing it out. It takes less space to write it as a row vector, so it is $(u_1, u_2)(v_1, v_2)^T$. However we write it, it often denoted $u \cdot v$ and called the *dot product*. I like this notation, as then we don't have to worry about whether we are writing u as a row or a column. I'll try to use this more. The book doesn't though.

Theorem 3.3. *A set of non-zero vectors that are pairwise orthogonal, are independent.*

Proof. Let $v_1, \ldots, v_n$ be pairwise orthogonal, non-zero vectors. To see they are independent, let $0 = c_1 v_1 + c_2 v_2 + \cdots + c_n v_n$. Taking inner products with v_i we get that

$$\begin{aligned} 0 = v_i \cdot 0 &= v_i \cdot (c_1 v_1 + c_2 v_2 + \cdots + c_n v_n) \\ &= c_1 v_i \cdot v_1 + c_2 v_i \cdot v_2 + \cdots + c_n v_i^t v_n \\ &= c_i v_i \cdot v_i = c_i \end{aligned}$$

This shows that $c_i = 0$, and works for all i, so the vectors are linearly independent. □

Orthogonal Subspaces

Definition 3.4. Two subspaces U and V of $\mathbb{R}^n$ are orthgonal, written $U \perp V$, if for all $u \in U$ and $v \in V$, $u \perp v$. For a subspace V of $\mathbb{R}^n$, the orthogonal complement $V^\perp$ (read 'V perp') of V, is the space of all vectors orthogonal to all vectors of V.

Note

To show that two subspaces are orthogonal, it is enough to show that every vector in a basis of one space is orthogonal to every vector in a basis of the other.

Observe that each row r in M and each vector x of the nullspace, $r \cdot x = 0$. As the rows of a matrix contain a basis of its rowspace we get that the row space and nullspace of any matrix are othogonal. In fact, as the nullspace is the set of all vectors x such that $r \cdot x = 0$, the nullspace $N(M)$ of a matrix M is exactly the orthogonal complement $C(M^T)^\perp$ of its row space.

Example 3.5. The matrix $M = \left[\begin{smallmatrix} 1 & 2 & 1 \\ 2 & 1 & 5 \end{smallmatrix}\right] \rightsquigarrow \left[\begin{smallmatrix} 1 & 0 & 3 \\ 0 & 1 & -1 \end{smallmatrix}\right]$ has rowspace $\{a(1,0,3) + b(0,1,-1) \mid a, b \in \mathbb{R}\}$ and nullspace $\{z(-3,1,1) \mid z \in \mathbb{R}\}$. We have

$$(1,0,3) \cdot (-3,1,1) = 0 \qquad \text{and} \qquad (0,1,-1) \cdot (-3,1,1) = 0,$$

as expected.

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Now consider a subspace V of $\mathbb{R}^n$ with basis $b_1, \ldots, b_r$, and let M be the matrix whose rows are the b_i. Then $V^\perp = N(M)$, which gives us $\dim(V^\perp) = n - r$, and so $\dim((V^\perp)^\perp) = n - (n - r) = r$. As we clearly have $V \subseteq (V^\perp)^\perp$, we get that $V = (V^\perp)^\perp$.

Fact 3.6. *For any subspace $V \subseteq \mathbb{R}^n$, $(V^\perp)^\perp = V$ and $\dim(V^\perp) = n - \dim(V)$.*

Applying this to a matrix M, and its transpose, we get the following.

Theorem 3.7. *[Fundamental Theorem of Linear Algebra] For an $m \times n$ matrix M,*

- *$C(M^T)^\perp = N(M)$ and $N(M)^\perp = C(M^T)$, and*
- *$C(M)^\perp = N(M^T)$ and $N(M^T)^\perp = C(M)$.*

Practice

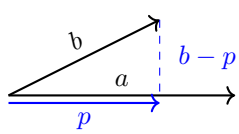
What is the dimension of the orthogonal complement $V^\perp$ of a space V spanned by the vectors $(1, 2, 4)$ and $(2, 1, 1)$? Find $V^\perp$?

Problems from the text

3.1: 3, 6, 7, 8, 12, 15, 17, 23, 35 (6, 4, 1, 19, 14, 20, 18, 26, 42)

3.2 Cosines and Projections

We talked about normalising vector spaces– changing the basis so that it consists of unit vectors. This was easy. We now look at changing the basis to a basis of pairwise orthogonal vectors. The idea is simple, for for two basis vectors a and b we split b up into an a-part p and a part $b - p$ that is orthogonal to a.



We can then write b as $p + (b - p)$ so $\{a, b - p\}$ spans the same space as $\{a, b\}$. This p such that $b - p$ is orthogonal to a is called the *projection* of b onto a, and denoted $p = \text{proj}_a(b)$. It is the vector of length $\|b\| \cos\theta$ in the direction of a so we can write it as

$$\text{proj}_a(b) = \|b\| \cos\theta \frac{a}{\|a\|}. \tag{1}$$

We get a nicer formula for this using the inner product.

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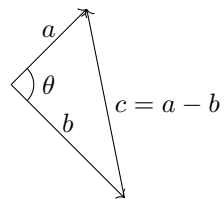
Formulae for $a \cdot b$ and for $\text{proj}_a(b)$

We saw that $a \perp b$ if and only if $a^T b = 0$. Let's start with an expression for $a^T b$ when $a \not\perp b$. Recall the pythagorean theorem for non-right angles (i.e., the cosine law):



$$c^2 = a^2 + b^2 - 2ab\cos\theta$$

On the one hand, we have

$$\begin{aligned} \|a-b\|^2 &= (a-b)\cdot(a-b) = a\cdot a + b\cdot b - b\cdot a - a\cdot b \\ &= a\cdot a + b\cdot b - 2a\cdot b \end{aligned}$$

while on the other hand we have by the cosine rule

$$\begin{aligned} \|a-b\|^2 &= \|a\|^2 + \|b\|^2 - 2\|a\|\cdot\|b\|\cos\theta \\ &= a\cdot a + b\cdot b - 2\|a\|\cdot\|b\|\cos\theta \end{aligned}$$

From this we see that $a \cdot b = \|a\| \cdot \|b\| \cos\theta$ where θ is the angle between a and b. Solving this for $\|b\|\cos\theta$ and plugging it into (1) we get

$$\text{proj}_a(b) = \frac{a\cdot b}{\|a\|}\frac{a}{\|a\|} = \frac{a\cdot b}{a\cdot a}a.$$

Memory Aid:

$$\cos\theta = \frac{u}{\|u\|}\cdot\frac{v}{\|v\|}$$

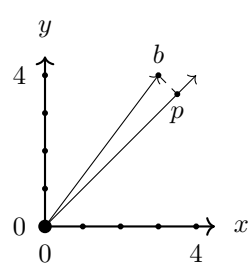
or if u and v are unit, just

$$\cos\theta = u\cdot v$$

We use the projection formula to:

- find the distance from a point to a line,
- write a vector in terms of a basis, or
- find an orthanormal basis of a space.

Example 3.8. To find the projection p of the point $b = (3, 4)$ onto the line $y = x$



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we project $b = (3,4)$ as a vector onto the vector $a = (1,1)$ which defines the line. So p is $\frac{b \cdot a}{a \cdot a} a = \frac{(3,4)\cdot(1,1)}{2} a = \frac{7}{2}(1,1)$.

We often refer to the difference $e = b - \text{proj}_a(b) = (3,4) - (3.5, 3.5) = \frac{1}{2}(-1,1)$ as the *error*. Its length $\|e\| = \sqrt{2}/2$ is the distance from p to $y = x$.

Example 3.9. We find an orthanormal basis of the subspace U of $\mathbb{R}^3$ that is generated by $\{a = (1,1,0), b = (1,2,3)\}$.

First, we orthogonalise by removing the a-component of b:

$$\begin{aligned} b' &= b - \text{proj}_a(b) = (1,2,3) - \frac{(1,2,3)\cdot(1,1,0)}{(1,1,0)\cdot(1,1,0)}(1,1,0) \\ &= (1,2,3) - \frac{3}{2}(1,1,0) = (-1/2, 1/2, 3). \end{aligned}$$

So $\{a, b'\}$ is an orthogonal basis of U. Now, normalising, we get the orthonormal basis $\{\sqrt{1/2}(1,1,0), \sqrt{1/38}(-1,1,6)\}$

Finding an orthanormal basis of a higher dimensional subspace is the same idea: from the third basis vector we have to remove the projections onto the previous two. We will come back to this in a couple of sections, and look at an orderly way to do it.

We have one last thing to point out here. The projection onto a vector a is a linear transformation, so can be represented as (multiplication on the left by) a matrix P. Indeed, as $a^T b$ is a scalar, we have $(a^T b)a = a(a^T b)$ so the projection formula becomes

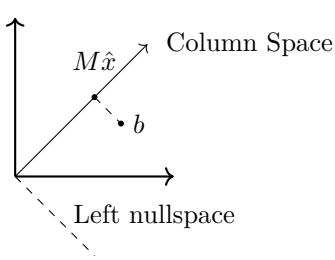
$$\text{proj}_a(b) = \frac{a^T b}{a^T a} a = \frac{a a^T}{a^T a} b =: Pb,$$

where $\frac{1}{a^T a}$ is a scalar and aa^T is a square matrix, so $P := \frac{aa^T}{a^T a}$ is a square matrix.

Problems from the text

3.2: 5, 9, 11, 12, 15, 19, 20, 22 (11, 12, 3, 10, 16, 17, 19, 24)

3.3 Projections and least squares



Sometimes the matrix equation $Mx = b$ we are trying to solve has no solution, but, entitled as we are, we want one anyways. That there is no solution means that b is not in the column space. We settle for the vector in the column space that is closest to b. That is, the projection of b onto the column space C. We find $\hat{x}$ such that $M\hat{x} = \text{proj}_C(b)$. The error $e = b - M\hat{x}$, is

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orthogonal to the column space, so is in the left nullspace of M.

So we would like to find $M\hat{x}$ and $\hat{x}$. How do we do this? Observe first that as $M\hat{x} = b - e$ we have

$$M^T M\hat{x} = M^T(b - e) = M^T b - 0 = M^T b.$$

So $\hat{x}$ is the solution to the related equation

$$(M^T M)x = M^T b.$$

Now while M is not invertible (because $Mx = b$ has no solution), $M^T M$ is (or at least we can make it so by removing any linearly dependent columns). Multipling on the left by $(M^T M)^{-1}$ we get:

$$\hat{x} = (M^T M)^{-1} M^T b \qquad \text{and so} \qquad M\hat{x} = M(M^T M)^{-1} M^T b.$$

We don't divide by matrices, but if we did, we might write the projection matrix at $\frac{MM^T}{M^T M}$. Compare this to the formula for the projection matrix projecting onto a vector a.

This $\hat{x}$ is called the *least squares estimate* for $Mx = b$. We will explain why in a second. As $M\hat{x}$ is the projection of x onto the column space of M, this matrix $M(M^T M)^{-1} M^T$ is called the *projection matrix*. The matrix $M^T M$ is called the *cross product matrix*.

We will look at some properties of these matrices. But first lets practice computing with them.

Example 3.10. It is clear by looking at the first and third row that when $M = \begin{bmatrix} -1 & -1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix}$, the equation $Mx = b$ has no solution. We find $\hat{x}$ and $M\hat{x}$.

Indeed,

- $M^T M = \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}$
- $(M^T M)^{-1} : \left[\begin{array}{cc|cc} 2 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right] \rightsquigarrow \left[\begin{array}{cc|cc} 1 & 0 & 3/2 & -1 \\ 0 & 1 & -1 & 1 \end{array}\right]$
- $\hat{x} = (M^T M)^{-1} M^T b = \begin{bmatrix} 3/2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 3/2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -7/2 \\ 3 \end{bmatrix}$
- $M\hat{x} = \begin{bmatrix} -1 & -1 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} -7/2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 3 \\ -1/2 \end{bmatrix}$

Properties of the cross product and projection matrices

For an $m \times n$ matrix M the cross product matrix $M^T M$ is a symmetric $n \times n$ matrix which is invertible if and only if the rank of M is n. Indeed, it has rank n if and only if M does. It cannot have greater rank than M as each row is a linear combination of rows of M. On the other hand, it cannot have rank less than that of M because of the following.

Practice

Show that M and $M^T M$ have the same nullspace.

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The projection matrix $P = P_M = M(M^T M)^{-1} M^T$ projects any vector b onto the column space $C(M)$ of M. That is to say, for all $b \in \mathbb{R}^n$

- $Pb \in C(M)$, and
- $e = b - Pb$ is orthogonal to $C(M)$.

We say that a matrix P is a *projection matrix* if there is any matrix M for which these properties hold.

Practice

Show that P_M is symmetric and idempotent: $P^2 = P$.

Now, we claim that any matrix P that is symmetric and idempotent is a projection matrix. Indeed, assume that P is symmetric and idempotent. P certainly maps b to $C(P)$. To see that $b - Pb$ is orthogonal to $C(P)$ we show that it is orthogonal to any $Pc \in C(P)$:

$$(b - Pb)^T Pc = b^T (I - P)^T Pc = b^T (P - P^2)c = 0.$$

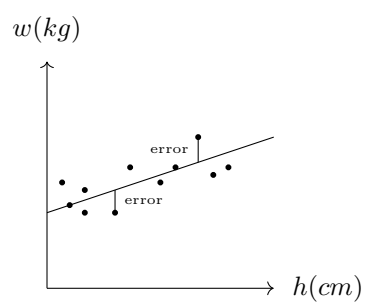
An application: the least squares line

A nice application of finding the approximate solution $\hat{x}$ of a system is using it to find the least squares regression line of a set of data.

We believe that a persons weight w is approximately a linear function of their height h. That is, we believe that

$$w = mh + b$$

for some constants m and b. We sample n people in order to find m and b.



The points seem to best fit some line $w = mh + b$. But what line? What are m and b? Ideally we want m and b such that for each point (w_i, h_i) we have $w_i = mh_i + b$. We want to solve $Mx = w$, i.e.,

$$\begin{bmatrix} 1 & h_1 \\ 1 & h_2 \\ \vdots & \\ 1 & h_n \end{bmatrix} \begin{bmatrix} b \\ m \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix},$$

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but it has no solution. How about replacing w with the closest vector $\hat{w}$ in the column space, and finding the solution $\hat{x}$ for that. What this does is minimises $\|MX - w\|$ and so

$$\|Mx - w\|^2 = \sum_i (b + mh_i - w_i)^2 = \sum_i e_i^2.$$

So this is called the least squares line.

Example 3.11. What is the the least squares line for the points $(1,3),(1,4)$ and $(2,6)$. We get the approximate solution to:

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 2 \end{bmatrix}}_{M} x = \underbrace{\begin{bmatrix} 3 \\ 4 \\ 6 \end{bmatrix}}_{b}.$$

- $M^T M = \left[\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \end{smallmatrix}\right] \left[\begin{smallmatrix} 1 & 1 \\ 1 & 1 \\ 1 & 2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 3 & 4 \\ 4 & 6 \end{smallmatrix}\right]$
- $(M^T M)^{-1} = \frac{1}{2}\left[\begin{smallmatrix} 6 & -4 \\ -4 & 3 \end{smallmatrix}\right]$
- $M^T b = \left[\begin{smallmatrix} 1 & 1 & 1 \\ 1 & 1 & 2 \end{smallmatrix}\right] \left[\begin{smallmatrix} 3 \\ 4 \\ 6 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 13 \\ 19 \end{smallmatrix}\right]$
- $\hat{x} = -\frac{1}{2}\left[\begin{smallmatrix} 6 & -4 \\ -4 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 13 \\ 19 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 \\ 5/2 \end{smallmatrix}\right]$

So $w = \frac{5}{2}h + 1$ is our least squares line.

Problems from the text

3.3: 1, 3, 6, 7, 8, 9, 11, 15, 17, 18, 22 (5, 2, 6, 10, 8, 13, 14, 12, 11, 18, 20)

3.4 Orthogonal Basis and Gram Schmidt

Given a basis $\{b_1, \ldots, b_m\}$ of a subspace V of $\mathbb{R}^n$ we now find an equivalent orthanormal basis $\{q_1, \ldots, q_m\}$ using the following 'Gram-Schmidt process'. First, let $q_1 = b_1/\|b_1\|$ and for $i = 2, \ldots, m$ let $q_i = q_i'/\|q_i'\|$ where

$$q_i' = b_i - \sum_{j=1}^{i-1} \operatorname{proj}_{q_j} b_i.$$

Example 3.12. Let $V \subseteq \mathbb{R}^3$ be generated by

$$B = \{b_1 = (1,0,1), b_2 = (1,0,0), b_3 = (2,1,0)\}.$$

Clearly the standard normal basis is an equivalent basis, but ignoring this for the moment, we use the Gram-Schmidt process to get another orthanormal basis.

- $q_1 = b_1/\|b_1\| = \frac{1}{\sqrt{2}}(1,0,1)$

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- $q_2' = b_2 - (q_1^T b_2)q_1 = (1,0,0) - \frac{1}{\sqrt{2}}(1)\frac{1}{\sqrt{2}}(1,0,1) = (1/2, 0, -1/2)$
- $q_2 = \frac{1}{\sqrt{2}}(1,0,-1)$
- $q_3' = b_3 - (q_1^T b_3)q_1 - (q_2^T b_3)q_2 = \cdots = (0,1,0)$
- $q_3 = (0,1,0)$.

Decomposing a vector with an orthonormal basis

Okay. That was fun, but what is really fun is expressing a vector as a linear combination of basis vectors. Once we have an orthanormal basis, this is easy. Indeed, any vector $v \in V$ can be written as

$$v = r_1 q_1 + r_2 q_2 + \cdots + r_m q_m,$$

for some scalars r_i. But as $q_i^T v = \sum_{j=1}^m r_j q_i^T q_j = r_i$ this can be written

$$v = \sum_{i=1}^m (q_i^T v) q_i = \sum_{i=1}^m q_i (q_i^T v),$$

or more compactly as

$$v = QQ^T v$$

where Q is the matrix whose i^{th} column is q_i so

$$Q^T v = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_m \end{bmatrix}$$

Properties of Orthogonal Matrices

The matrix Q whose columns are a set of orthanormal vectors has some other nice properties that arise from the fact that $Q^T Q = I$.

Such a matrix Q **preserves inner products**

$$(Qu)^T (Qv) = u^T Q^T Q v^T = u^T v.$$

It follows immediately that it **preserves length**:

$$\|Qv\| = \|v\|$$

and **preserves the angle between vectors**:

$$\cos^{-1}\left(\frac{(Qu)^T(Qv)}{\|Qu\| \cdot \|Qv\|}\right) = \cos^{-1}\left(\frac{u^T v}{\|u\| \cdot \|v\|}\right).$$

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As $Q^TQ = I$, we have that Q^T is the left inverse of Q. In the special case that Q is square, the matrix is called *orthogonal*, and $Q^T = Q^{-1}$, which is pretty convenient: as the inverse is immediate it is easy to solve $Qx = b$.

Even when Q is not square, the matrix QQ^T is square and has some nice properties. Indeed it is idempotent:

$$Q(Q^TQ)Q^T = QIQ^T = QQ^T.$$

So $QQ^T =: P_Q$ is the matrix that projects onto the column space of Q.

Thus the projection of b onto $C(Q)$ is QQ^Tb, and the least squares approximation for

$$Qx = b$$

is just $\hat{x} = Q^Tb$.

The QR-factorisation of M

Writing out our original basis $\{b_1, \ldots, b_m\}$ in terms of our orthonormal basis $\{q_1, \ldots, q_m\}$:

$$\begin{array}{lclllcllll}
b_1 & = & q_1^Tb_1q_1 & +q_2^Tb_1q_2 & +q_3^Tb_1q_3 & = & q_1^Tb_1q_1 & +0 & +0 \\
b_2 & = & q_1^Tb_2q_1 & +q_2^Tb_2q_2 & +q_3^Tb_2q_3 & = & q_1^Tb_2q_1 & +q_2^Tb_2q_2 & +0 \\
b_3 & = & q_1^Tb_3q_1 & +q_2^Tb_3q_2 & +q_3^Tb_3q_3 & = & q_1^Tb_3q_1 & +q_2^Tb_3q_2 & +q_3^Tb_3q_3
\end{array}$$

we see that this can be written as

$$\underbrace{\begin{bmatrix} b_1 \mid b_2 \mid b_3 \end{bmatrix}}_{M} = \underbrace{\begin{bmatrix} q_1 \mid q_2 \mid q_3 \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} q_1^Tb_1 & q_1^Tb_2 & q_1^Tb_3 \\ 0 & q_2^Tb_2 & q_2^Tb_3 \\ 0 & 0 & q_2^Tb_3 \end{bmatrix}}_{R}$$

where R is upper triangular.

To find the least squares solution $\hat{x}$ of the equation $Mx = b$ we had to find the solution to the equation

$$M^TMx = M^Tb.$$

Using a QR decomposition of M, $M^Tb = M^TMx$ becomes (working from the middle)

$$R^TQ^Tb = (QR)^Tb = \mathbf{M^Tb} = \mathbf{M^TMx} = (QR)^T(QR)x = R^T(Q^TQ)Rx = R^TRx.$$

Now as R is triangular, and invertible (as long as M has independent columns), so multiplying on the left by its inverse we get

$$Rx = Q^Tb.$$

To find $\hat{x}$ it is then enough to solve this equation, with triangular R. So having the QR decomposition of M makes finding the least squares solution of $Mx = b$ much faster.

Problems from the text

3.4: 1, 6, 7, 8, 10 (2, 14, 11, 10, 6)

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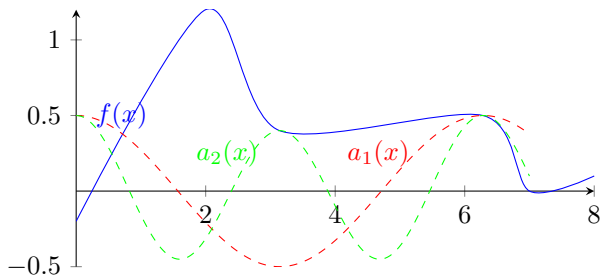
3.5 Fast Fourier Transforms

Take a continuous function $f(x)$. One can approximate it by any periodic function on some interval.



The idea is that the degree i approximation $a_i(x)$ of $f(x)$ by some periodic function p will take a linear combination

$$a_i(x) = c_1 p(x) + c_2 p(2x) + \cdots + c_i(p(ix))$$

of compressions of p so that $a_i(x)$ and $f(x)$ agree at i regularly spaced points in the period of p. The need not agree at other points, but because f is continuous, as the spaces gets smaller, even on off points in the interval, p gets close to f.

Let's see how this works with $p(x) = \cos(x)$. We will try to approximate $f(x)$ on the interval $[0, 2\pi]$; the ith approximation $a_i(x)$ will agree with $f(x)$ at the points $frac2\pi n\alpha$ for $\alpha = 1, \ldots, n$.

It is easy to see that to get $a_1(x) = c_1 \cos(x)$ to agree with $f(x)$ at $x - 2\pi$ we must take

$$a_1(x) = f(2\pi)\cos(x).$$

For $a_2(x)$ we want

$$a_2(x) = c_1 \cos(x) + c_2 \cos(2x)$$

to agree with $f(x)$ at π and 2π. Well, in compressing $\cos(x)$ to $\cos(2x)$ we've doubled its period. So both of these functions have the same value at 2π. Writing

$$a_2(x) = \frac{f(2\pi) + a}{2}\cos x + \frac{f(2\pi) - a}{2}\cos(2x)$$

for any constant a we still have $a_2(x)$ agreeing with $f(x)$ at $x = 2\pi-$ the constant a cancels. At the value $x = \pi$ however, the values of $\cos(x)$ and $\cos(2x)$ are opposite, so the values $f(2\pi)$ cancel and we get $-2a$. So letting $a = -f(\pi)/2$ we get that

$$a_2(x) = \frac{f(2\pi) - f(\pi)}{2}\cos x + \frac{f(2\pi) - f(\pi)}{2}\cos(2x)$$

agrees with $f(x)$ at π and 2π.

How can we make an approximation that agrees with $f(x)$ on 4, or n points? We can keep doing this, but lets first write this nonesense more compactly with some matrices.

To find $a_2(x)$ we really solved the matrix equation

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$$\begin{bmatrix} \cos(1\cdot\pi) & \cos(2\cdot\pi) \\ \cos(1\cdot 2\pi) & \cos(2\cdot 2\pi) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} f(\pi) \\ f(2\pi) \end{bmatrix}.$$

For this to have a solution it was important that the rows of

$$F_2 = \begin{bmatrix} \cos(1\cdot\pi) & \cos(2\cdot\pi) \\ \cos(1\cdot 2\pi) & \cos(2\cdot 2\pi) \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \tag{2}$$

were independent.

Practice

Solve

$$\begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} f(\pi) \\ f(2\pi) \end{bmatrix}.$$

Do you get the same c_1 and c_2 that I did?

To generalise to n points this with some periodic function $p(x)$ of period T we set $t = T/n$ and solve

$$\begin{bmatrix} p(t) & p(2t) & p(3t) & \dots & p(nt) \\ p(2t) & p(4t) & p(6t) & \dots & p(2nt) \\ & & \vdots & & \\ p(nt) & p(2nt) & p(3nt) & \dots & p(n^2t) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} f(t) \\ f(2t) \\ \vdots \\ f(nt) \end{bmatrix}.$$

The matrix $[F_n]_{ij} = p(t\cdot ij)$ is the nth Fourier matrix. When we are using the periodic function $\cos(x)$ with period $T = 2\pi$ we saw above that

$$F_2 = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix}.$$

We could solve it because it was invertible, but are we sure that F_n would be in general? Can we easily compute its inverse to solve for the c_i?

Well. There is a different function $p(x)$ for which these questions have nice clean answers. But it uses complex numbers.

Complex numbers

Recall that in the complex plane $\mathbb{C} = \mathbb{R} \times i\mathbb{R}$ a point $x + iy$ can be expressed as

$$re^{i\theta} = r(\cos\theta + i\sin\theta)$$

where θ is the angle between (x, y) and the x-axis, r is the length of (x, y), and $e^{i\theta} = \cos\theta + i\sin\theta$ is the complex exponential function.

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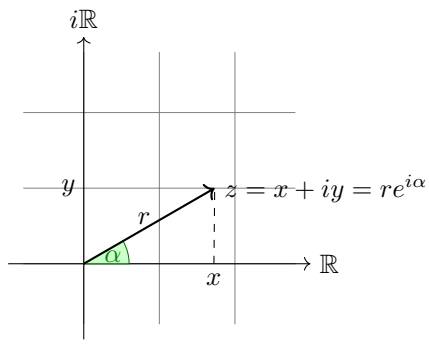

This notation is exceptionally useful, and is appropriate, as one can show that

$$e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')}.$$

What this means is that for points on the unit circle in the complex plane, we can multiply two points by simply adding their angles from the x-axis.

The number $\omega_n = e^{i\frac{2\pi}{n}}$ is the *primitive* n^{th} *root of unity*, as one has that

$$\omega_n^n = (e^{i\frac{2\pi}{n}})^n = e^{i2\pi} = 1.$$

So, for example $\omega_2 = -1$ and $\omega_4 = i$.

The have easy arithmetic:

$$\omega_n^a \cdot \omega_n^b = \omega_n^{a+b}$$

Practice

Show that $\omega_n^d = \omega_n^m$ where $m = d \mod n$.

The Fourier Matrix

Using $p(x) = e^{ix}$ which has period $T = 2\pi$ the ith row of F_n becomes

$$(\omega_n^i, \omega_n^{2i}, \ldots, \omega_n^{ni}).$$

We claim that these rows are orthogonal, as long as n is even. Indeed,

$$(\omega^i, \omega^{2i}, \ldots, \omega^{ni}) \cdot (\omega^j, \omega^{2j}, \ldots, \omega^{nj}) = \omega^{i+j} + \omega^{2(i+j)} + \cdots + \omega^{n(i+j)}.$$

This sum is a sum of even n vectors evenly spaced around the unit circle in the complex plane. The ith summand and the $(i + n/2)^{th}$ cancel out, so this sum is 0.

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We make the Fourier matrix from these rows. That is, for even n we let F_n be defined by

$$[F_n]_{ij} = \omega_n^{ij}.$$

For example,

$$F_2 = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \qquad F_4 = \begin{bmatrix} \omega_4 & \omega_4^2 & \omega_4^3 & \omega_4^4 \\ \omega_4^2 & \omega_4^4 & \omega_4^6 & \omega_4^8 \\ \omega_4^3 & \omega_4^6 & \omega_4^9 & \omega_4^{12} \\ \omega_4^4 & \omega_4^8 & \omega_4^{12} & \omega_4^{16} \end{bmatrix} = \begin{bmatrix} i & -1 & -i & 1 \\ -1 & 1 & -1 & 1 \\ -i & -1 & i & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

Now observe that

$$(\omega^i, \omega^{2i}, \ldots, \omega^{ni}) \cdot (\omega^{-i}, \omega^{-2i}, \ldots, \omega^{-ni}) = 1 + 1 + \cdots + 1 = n.$$

So where $\bar{F}_n$ is defined by $[\bar{F}_n]_{ij} = \omega_n^{-ij}$, we have $F_n \bar{F}_n = nI$. So $F_n^{-1} = \frac{1}{n}\bar{F}_n$.

For example

$$F_4^{-1} = \frac{1}{4}\begin{bmatrix} \omega_4^{-1} & \omega_4^{-2} & \omega_4^{-3} & \omega_4^{-4} \\ \omega_4^{-2} & \omega_4^{-4} & \omega_4^{-6} & \omega_4^{-8} \\ \omega_4^{-3} & \omega_4^{-6} & \omega_4^{-9} & \omega_4^{-12} \\ \omega_4^{-4} & \omega_4^{-8} & \omega_4^{-12} & \omega_4^{-16} \end{bmatrix} = \frac{1}{4}\begin{bmatrix} -i & -1 & i & 1 \\ -1 & 1 & -1 & 1 \\ i & -1 & -i & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}.$$

Example 3.13. We can approximate $f(x) = x^2$ on $[0, 2\pi]$ by computing

$$\frac{1}{4}\begin{bmatrix} -i & -1 & i & 1 \\ -1 & 1 & -1 & 1 \\ i & -1 & -i & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} (\pi/2)^2 \\ (2\pi/2)^2 \\ (3\pi/2)^2 \\ (4\pi/2)^2 \end{bmatrix} \approx \frac{\pi^2}{4}\begin{bmatrix} 3+2i \\ 5/2 \\ 3-2i \\ 15/2 \end{bmatrix}.$$

The function

$$\frac{\pi^2}{4}(3+2i)e^{i\frac{\pi}{2}} + 5/2e^{i2\frac{\pi}{2}} + (3-2i)e^{i3\frac{\pi}{2}} + 15/2e^{i4\frac{\pi}{2}}$$

is 2π periodic and agrees with x^2 at $\pi/2, \pi, 3\pi/2$ and 2π.

Problems from the text

3.5: 1,3,5,8,10,11

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4 Determinants

4.1 Introduction

The determinant of a 2×2 matrix $M = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ is

$$\det(M) = |M| = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

It is a number with a lot of nice properties. For example, $|M|$ is the volume of the box whose edges are the rows of M. Also $|M| = 0$ if and only if M is singular. If $|M| \neq 0$ then

$$M^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} / |M|.$$

In this chapter, we define the determinant for all square matrices and investigate its properties.

4.2 Properties of the Determinant

We have not defined the determinant $\det(M)$ of a square matrix M, except for when M is 2×2. In general, the determinant is a function from the set of $n \times n$ matrices to $\mathbb{R}$ such that the following three properties hold.

1) $\det(I) = \left|\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{smallmatrix}\right| = 1.$

2) A row exchange of M changes the determinant of M by a factor of -1.

3) The determinant is linear in the first row:

$$\begin{vmatrix} tw_1 & tw_2 & tw_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = t \begin{vmatrix} w_1 & w_1 & w_1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} \qquad \text{and} \qquad \begin{vmatrix} a_1+b_1 & a_2+b_2 & a_3+b_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix} + \begin{vmatrix} b_1 & b_2 & b_3 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}$$

Notice that the determinant is **not** linear in the whole matrix, just in the first row.

Practice

Using property 2) show that the determinant is linear in any row of the matrix. Show in particular that if we get M' by multiplying a row of M by r, then $|M'| = r|M|$.

Practice

Show that the above three properties hold for the determinant we defined for 2×2 matrices.

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The above three properties are enough to uniquely define the determinant. We now look at some properties that will help us compute it.

4) If two rows of M are equal, then $\det(M) = 0$.

5) The row operation of adding a multiple of one row to another doesn't change the determinat.

6) If M has a row of zeros, then $\det(M) = 0$.

Practice

We now know how the determinant acts under row operations, so we can compute the determinant using Gaussian elimination. Try this with the matrix $M = \left[\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 2 & 2 \\ 0 & 0 & 1 \end{smallmatrix}\right]$.

7) If M is triangular then $\det(M)$ is the product of the diagonal entries.

8) M is singular if and only if $|M| = 0$.

9) The determinant is multiplicative: $|AB| = |A||B|$.

10) $|M| = |M^T|$.

Practice

Prove properties 4) - 10) using properties 1) - 3). Hint: if 7,9 and 10) seems hard, make sure do the previous practice problem first.

Problems from the text

4.2: 2, 4, 5, 7, 8, 9, 17, 24, 25, 26 (1, 4, 7, 5, 10, 6, 19, 27, 22, 28)

4.3 Formulae for Determinants

We saw how to calculate the determinant by elimination, keeping track of row swaps:

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \\ 1 & 0 & 2 \end{vmatrix} = \begin{vmatrix} 1 & 2 & 3 \\ 0 & 0 & -2 \\ 0 & -2 & -1 \end{vmatrix} = -\begin{vmatrix} 1 & 2 & 3 \\ 0 & -2 & -1 \\ 0 & 0 & -2 \end{vmatrix} = -(1 \cdot -2 \cdot -2) = -4.$$

While this always works, there are other methods that are useful at times. For a 2×2 matrix $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ we can use the elimination method to get

$$\left|\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right| = \left|\begin{smallmatrix} a & b \\ 0 & d - b\frac{c}{a} \end{smallmatrix}\right| = ad - cb.$$

This is an easy formula to memorise, and quick to use.

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Practice

Compute the determinant of $\left[\begin{smallmatrix} 1 & 4 \\ 3 & 1 \end{smallmatrix}\right]$.

We can derive this same formula by linearity:

$$\begin{aligned}
\left|\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right| &= \left|\begin{smallmatrix} a & 0 \\ c & d \end{smallmatrix}\right| + \left|\begin{smallmatrix} 0 & b \\ c & d \end{smallmatrix}\right| \\
&= \left|\begin{smallmatrix} a & 0 \\ c & 0 \end{smallmatrix}\right| + \left|\begin{smallmatrix} a & 0 \\ 0 & d \end{smallmatrix}\right| + \left|\begin{smallmatrix} 0 & b \\ c & 0 \end{smallmatrix}\right| + \left|\begin{smallmatrix} 0 & b \\ 0 & d \end{smallmatrix}\right| \\
&= 0 + ad\left|\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right| + bc\left|\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right| + 0 \\
&= ad - bc
\end{aligned}$$

What have we done? We have decomposed the matrix into 'summand matrics' with one non-zero entry in each row. Unless there was one non-zero entry in each column, the determinant of the summand was zero. If there was one non-zero entry per column, their determinant was the product of the entries, times a factor of -1 if the columns were not in the right order.

We can do the same for an $n \times n$ matrix. In this decompositon, we get a non-zero summand by choosing a permuation of the columns, and then multiplying the entries down the diagonal. We sum these summands, multiplied by -1 depending on the number of column switches we used.

Observe that for a permutation matrix

$$\begin{vmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = -\begin{vmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1.$$

That is, the determinant is $(-1)^s$ if it takes s switches to get to the identity. This yields the *permutation formula for the determinant*:

$$|M| = \sum_P \text{Product of diagonals of } PM \cdot |P|$$

where the sum is over all $n!$ permutation matrices P.

This notation is a bit clumsy. For a permuation matrix P let the *permutation of* P be the vector $(\sigma_P(1), \sigma_P(2), \ldots \sigma_P(n))$ be such that the ith column of P is the $\sigma_P(i)^{th}$ column of I.

Example 4.1. The permuation of $P = \left[\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{smallmatrix}\right]$ is $(1, 3, 2)$.

Where S_m is the set of $n \times n$ permutation matrices, the permuation formula for $M = [m_{i,j}]$ then becomes

$$\det(M) = \sum_{P \in S_n} (\det(P) \prod_{i=1}^{n} m_{i\sigma_P(i)}).$$

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Example 4.2.

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 4 & 4 \\ 1 & 0 & 2 \end{vmatrix} = \begin{matrix} 1 & 2 & 3 \\ 2 & 4 & 4 \\ 1 & 0 & 2 \end{matrix} - \begin{matrix} 1 & 3 & 2 \\ 2 & 4 & 4 \\ 1 & 2 & 0 \end{matrix} - \begin{matrix} 2 & 1 & 3 \\ 4 & 2 & 4 \\ 0 & 1 & 2 \end{matrix} + \begin{matrix} 2 & 3 & 1 \\ 4 & 4 & 2 \\ 0 & 2 & 1 \end{matrix} - \begin{matrix} 3 & 2 & 1 \\ 4 & 4 & 2 \\ 2 & 1 & 0 \end{matrix} + \begin{matrix} 3 & 1 & 2 \\ 4 & 2 & 4 \\ 2 & 1 & 0 \end{matrix}$$

$$= 8 - 0 - 8 + 8 - 12 + 0 = -4$$

This permutation formula can be described in another way, it leads to what is called co-factor expansion formula.

The $(i,j)^{th}$ *minor* of an $n \times n$ matrix M is the $(n-1) \times (n-1)$ matrix M_{ij} that we get from M by removing the ith row and the jth column. The $(i,j)^{th}$ *cofactor* of M is $c_{ij} = (-1)^{i+j}|M_{i,j}|$.

The cofactor formula says that fixing i we get

$$|M| = \sum_{j=1}^{n} c_{ij} m_{ij}$$

or fixing j we get

$$|M| = \sum_{i=1}^{n} c_{ij} m_{ij}.$$

The first equation is called the 'cofactor expansion along the ith row' and the second is the 'cofactor expansion along the jth column'.

Example 4.3. Expanding along the first row we get:

$$\begin{aligned} \begin{vmatrix} 2 & 3 & 4 \\ 1 & 0 & 3 \\ -1 & 3 & 2 \end{vmatrix} &= 2\begin{vmatrix} 0 & 3 \\ 3 & 2 \end{vmatrix} - 3\begin{vmatrix} 1 & 3 \\ -1 & 2 \end{vmatrix} + 4\begin{vmatrix} 1 & 0 \\ -1 & 3 \end{vmatrix} \\ &= 2(-7) - 3(5) + 4(3) = -17 \end{aligned}$$

Practice

Try expanding along the second column.

Practice

True or false:

- The determinant of $S^{-1}MS$ is the same as that of M.
- If $\det(M) = 0$ then at least one of the cofactors of M is 0.
- A matrix whose entries are 0 and 1 has determinant $-1, 0$ or 1.

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Practice

Show that

$$\begin{vmatrix} * & * & * & * & * \\ * & * & * & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & * & * \end{vmatrix} = 0$$

where each $*$ is a different real number.

Problems from the text

4.3: 1,3,10,11,13(A,B),15,23,27 (3, 1, 8, 7, 14(A,B), 18, 17, 25)

4.4 Applications of Determinants

A formula for M^{-1}

Observe that for a 2×2 matrix $M = \left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ we have that

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \det(M) I$$

and so we get the formula

$$M^{-1} = \frac{1}{\det M} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Practice

Compute $\begin{vmatrix} 3 & 8 \\ 3 & -1 \end{vmatrix}$.

We can generalise this formula. Recalling that the cofactor c_{ij} of M is $(-1)^{i+j}|M_{ij}|$, the cofactor matrix C of M is $C = [c_{ij}]$.

The cofactor matirx of $\left[\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right]$ is

$$C = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}.$$

So this equation above becomes

$$M^{-1} = C^T / \det M. \tag{3}$$

This holds not only for 2×2 matrices, but for all square matrices. Indeed, consider the product

$$MC^T = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} c_{11} & c_{21} & \dots & c_{n1} \\ \vdots & & & \vdots \\ c_{1n} & c_{2n} & \dots & c_{nn} \end{bmatrix}.$$

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The diagonal entries of the product are $\det(M)$ by the co-factor expansion formula. The non-diagonal entries are 0 as, say, the entry in the first row and second column is

$$a_{11}c_{21} + a_{12}c_{22} + a_{13}c_{23} + \cdots + a_{1n}c_{2n}.$$

Now what is this? This is the expansion along the first row of a matrix with the same first row of as M, but we are replacing the cofactor c_{1n} with the cofactor c_{2n}. This is the $(i,n)^{th}$ cofactor of the matrix M' that we get from M by replacing its second row with its first. So the whole expression is the determinant of the matrix M', the first two rows of which are the first row of M. Thus the expression is 0.

We thus have

$$MC^T = \det(M)I$$

which proves the formula.

Example 4.4.

$$\begin{bmatrix} 1 & 3 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 3 \end{bmatrix}^{-1} = \frac{1}{3}\begin{bmatrix} 3 & 0 & 0 \\ -9 & 3 & 0 \\ 10 & -4 & 1 \end{bmatrix}^T = \frac{1}{3}\begin{bmatrix} 3 & -9 & 10 \\ 0 & 3 & -4 \\ 0 & 0 & 1 \end{bmatrix}$$

Using this formula isn't always quicker than simply eliminating to a traingular matrix, but sometimes it is.

Cramer's Rule

If $Mx = b$, then $x = M^{-1}b = \frac{1}{\det M}C^T b$. Reading just the ith row of this, we get that

$$x_i = \frac{b_1c_{i1} + b_2c_{i2} + \cdots + b_nc_{in}}{\det M}.$$

This is the cofactor expansion along the ith column of the matrix B_i which we get from M by replacing the ith column with the vector b. So we get

$$x_i = \det B_i / detM.$$

This is called Cramer's Rule.

Example 4.5. To solve $\left[\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right]x = \left[\begin{smallmatrix} 0 \\ 6 \end{smallmatrix}\right]$ by Cramer's rule, we can compute that

$$x_1 = \frac{\left|\begin{smallmatrix} 0 & 3 \\ 6 & 4 \end{smallmatrix}\right|}{\left|\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right|} = \frac{-18}{-2} = 9$$

$$x_2 = \frac{\left|\begin{smallmatrix} 1 & 0 \\ 2 & 6 \end{smallmatrix}\right|}{\left|\begin{smallmatrix} 1 & 3 \\ 2 & 4 \end{smallmatrix}\right|} = \frac{6}{-2} = -3$$

so $x = (9, -3)$.

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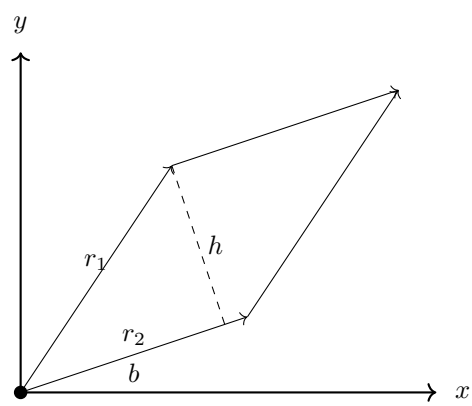
Volume of a box

Given a matrix M with orthogonal rows $b_1, b_2, \ldots, b_n$ we have that

$$MM^T = \begin{bmatrix} \|b_1\|^2 & 0 & \ldots & 0 \\ 0 & \|b_2\|^2 & \ldots & 0 \\ \vdots & & & \vdots \\ 0 & \ldots & 0 & \|b_n\|^2 \end{bmatrix},$$

so $\det(MM^T) = \prod \|b_i\|^2 = \prod l_i^2$ where l_i is the length of b_i.

But $\det(MM^T) = \det(M)\det(M^T) = \det(M)^2$ and so $|\det(M)| = \prod l_i$ is the volume of the box whose edges are the vectors b_i. The most wonderful thing you have seen in days, is that this continues to hold when the sides of the box are not orthogonal. Indeed, consider the parallelopiped shown below with edges r_1 and r_2. It has volume $b \cdot h$, where $b = \|r_2\|$ and $h = \|r_1 - \frac{r_1^T r_2}{r_1^T r_1} r_1\|$.



We saw that $\det(M)$ doesn't change when we remove a multiple of one row from another, so replacing r_1 with the vector h which is perpendicular to r_2, the determinant is $b \cdot h$ as needed. The same argument works in higher dimensions, so we get the following.

Fact 4.6. *Where M is the matrix whose row vectors are the edges of a parallelopiped P, $|\det(M)|$ is the volume of P.*

Practice

A box has edges from $(0,0,0)$ to $(3,1,1), (1,3,1)$ and $(1,1,3)$. Find its volume.

As the determinant of M is preserved by taking the transposea, we can alternately view it as the volume of the figure spanned by the columns of M. Viewing M as a transformation, it transforms the unit cube into this parallelopiped having volume $|\det(M)|$. More generally though, as a transformation, it transforms any solid in $\mathbb{R}^n$.

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Let B be the set of points that make up the ball of volume $\frac{4}{3}\pi$:

$$B = \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}.$$

Transforming by M maps this set to some other solid (an ellipsoid) that has volume $\frac{4}{3}\pi \cdot |\det M|$.

Finding Pivots

Consider eliminating a non-singular matrix M to triangular matrix T without scaling any rows or doing row switches. (So we only do row operations under which the determinant is unchanged.) If we can succesfully eliminate M to triangular T, let the ith diagonal entry of the eliminated matrix be called the ith pivot t_i. Computing determinants, we can find these pivots without eliminating.

Let M_m refer to the square matrix made up of the intersection of the first m rows and m columns of M. Observe that as we eliminate M to T, we are also eliminating M_m to T_m by the same determinant preserving operations, so for each m, we have $\det(T_m) = \det(M_m)$.

So clearly $M_1 = T_1$ has the single entry t_1. The second pivot t_2 is $|T_2|/|T_1| = |M_2|/|M_1|$, and inductively we see that

$$t_i = |M_i|/|M_{i-1}|$$

for all i.

Example 4.7. The pivots of $\begin{bmatrix} 3 & 1 & 4 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{bmatrix}$ are

$$t_1 = 3$$
$$t_2 = \frac{\left|\begin{smallmatrix} 3 & 1 \\ 1 & 2 \end{smallmatrix}\right|}{3} = \frac{5}{3}$$
$$t_3 = \frac{\left|\begin{smallmatrix} 3 & 1 & 4 \\ 1 & 2 & 3 \\ 1 & 3 & 5 \end{smallmatrix}\right|}{5} = \frac{5}{5} = 1$$

Wonderful. As we have to compute the full determinant in the last step, and this essentially requires triangulating the matrix, this seems like a painstaking way to compute these pivots. But is does give us some insight. We see the following fact that will come in handy.

Fact 4.8. *The matrix M can be eliminated to triangular without row exchanges if and only if the leading submatrices $M_1, M_2, \ldots, M_n$ are non-singular.*

Problems from the text

4.4: 1, 3(A), 4, 7, 8, 10, 14, 19, 23, 29, 32 (1, 7(A), 8, 3, 6, 11, 13, 20, 21, 28, 29)

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5 Eigenvectors and eigenvalues

5.1 Introduction

An *eigenvalue* of a matrix M is a constant λ such that

$$Mx = \lambda x$$

for some non-zero vector x. The vector x is the corresponding *eigenvector*.

Example 5.1. For the matrix $M = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]$, we have

$$M\left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 \\ 0 \end{smallmatrix}\right] \text{ and } M\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 0 \\ 3 \end{smallmatrix}\right] = 3\left[\begin{smallmatrix} 0 \\ 1 \end{smallmatrix}\right],$$

so 1 is an eigenvalue with eigenvector $(1, 0)$ and 3 is another eigenvalue with eigenvector $(0, 1)$. There are no more eigenvalues in this example. We will see soon that **an $n \times n$ matrix can have at most n eigenvalues.**

Practice

Does $M = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]$ does have more **eigenvectors**?

Indeed $M\left[\begin{smallmatrix} 7 \\ 0 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 7 \\ 0 \end{smallmatrix}\right]$. All vectors in the space $\langle(1, 0)\rangle = \{r(1, 0) \mid r \in R\}$ are eigenvectors with eigenvalue 1.

Practice

Show that the set of vectors that are eigenvectors for a given eigenvalue λ are a space.

This space is the *eigenspace* of λ.

So $M = \left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]$ has eigenvectors in the spaces $\langle(1, 0)\rangle$ and $\langle(0, 1)\rangle$.

Practice

Show that M has no other eigenvectors.

Practice

What are the eigenvectors of the rotation $M = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$?

In this example it was easy to find the eigenvectors by inspection. This is not always the case. What are the eigens (eigenvalues and eigenvectors) of $M = \left[\begin{smallmatrix} 2 & 3 \\ 1 & 4 \end{smallmatrix}\right]$? Well we want v such that $Mv = \lambda v$ for some λ. We can write this as the matrix equation $Mv = \lambda I v$, and re-arrange this to get

$$(M - \lambda I)v = 0.$$

So we are looking for λ such that the *characteristic matrix* $M - \lambda I$ of M is singular. Its nullspace is the eigenspace of λ. As a matrix is singular if and only if its determinant is 0, we can find the eigenvalue λ for M as follows.

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Example 5.2. To find the eigens of $M = \left[\begin{smallmatrix} 2 & 3 \\ 1 & 4 \end{smallmatrix}\right]$ we write out its *characteristic equation*:

$$0 = |M - \lambda I| = \begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix}.$$

Using our determinant formula to expand on the right to find the *characteristic polynomial* of M:

$$\begin{aligned} \begin{vmatrix} 2-\lambda & 3 \\ 1 & 4-\lambda \end{vmatrix} &= (2-\lambda)(4-\lambda) - 3 \\ &= \lambda^2 - 6\lambda + 5 \\ &= (\lambda - 1)(\lambda - 5) \end{aligned}$$

solve the quadratic equation $(\lambda - 1)(\lambda - 5) = 0$ to get $\lambda = 1$ or 5.

We usually write the eigenvalues in increasing order as $\lambda_1 = 1$ and $\lambda_2 = 5$.

Practice

Find the eigenspaces of $\lambda_1 = 1$ and $\lambda_2 = 5$ by finding the nullspaces of $M - \lambda_1 I$ and $M - \lambda_2 I$.

The same process works for any square matrix M:

- Find eigenvalues λ_i by solving the characteristic equation $0 = |M - \lambda I|$.
- Find the eigenspace of λ_i by finding the nullspace of $M - \lambda_i I$.

Practice

We asked earlier what the eigenvectors of the rotation matrix $R = \left[\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right]$ are? Likely you said that as it rotates all vectors by 90 degrees, and this not a scale, there are none, so no eigenvalues either. Use the characteristic polynomial of R to find its eigenvalues, (They exist!) and their eigenvectors.

When the matrix M has dimension greater than 3, the characteristic polynomial has degree greater than three, and finding its roots is usually hard to do. We will have to develop a different way to find the eigens for larger matrices. But before we go onto this, we point out a few larger matrices for which the above approach works, or for which we can use ad hoc reasoning.

Example 5.3. If M is singular, so the nullspace $N(M)$ is more than just $\{0\}$, any vector v in $N(M)$ is an eigenvector with eigenvalue 0. The dimension of the eigenspace for 0 is the nullity of M.

Example 5.4. If P is the projection matrix projecting on the space $V \subset R^n$, then all vectors in V are eigenvectors with eigenvalue 1. Any vector orthogonal to V is an eigenvector with eigenvalue 0.

Example 5.5. The eigenvalues of a triangular matrix are the diagonal entries. Indeed, if say $M = \left[\begin{smallmatrix} 1 & 1 & 2 & 3 \\ & 4 & 2 & 3 \\ & & -2 & 1 \\ & & & 4 \end{smallmatrix}\right]$ then

$$|M - \lambda I| = \left[\begin{smallmatrix} 1-\lambda & 1 & 2 & 3 \\ & 4-\lambda & 2 & 3 \\ & & -2-\lambda & 1 \\ & & & 4-\lambda \end{smallmatrix}\right] = (1-\lambda)(4-\lambda)(-2-\lambda)(4-\lambda),$$

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which has roots $1, 4, -2$ and 4.

In this previous example, 4 was a root of the characteristic equation of *multiplicity* two. As the characteristic polynomial of an $n \times n$ matrix has degree n there are exactly n eigenvalues, with multiplicity. (But some might not be real.) We will see that an eigenvalue having multiplicity d means that the corresponding eigenspace has dimension at most d.

As the eigenvalues of a triangular matrix are easy to determine, one might try to use Gaussian elimination to triangulise the matrix and then find the eigen values. Unfortunately, elimination does not preserve eigenvalues– if it did, then every non-singular matrix would have the same eigenvalues as the identity. We will look at a different kind of 'diagonalisation' that preserves eigenvalues.

Practice

Show that $|M|$ is the product of the eigenvalues of a matrix M. Show that the sum of the eigenvalues of M is the sum of the diagonal entries.

Problems from the text

5.1: 1, 3, 6, 7, 9, 10, 12, 17, 18, 24, 28 (Non International Student Edition: 1, 3, 8, 5, 11, 9, 13, 19, 20, 27, 26)

5.2 Diagonalising a matrix

Let S be an $n \times n$ matrix whose columns are eigenvectors of some $n \times n$ matrix M. We get

$$MS = M\left[\begin{array}{c|c|c|c} x_1 & x_2 & \dots & x_n \end{array}\right] = \left[\begin{array}{c|c|c|c} \lambda_1 x_1 & \lambda_2 x_2 & \dots & \lambda_n x_n \end{array}\right] = S\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \dots & \\ & & & \lambda_n \end{bmatrix},$$

where λ_i is the eigenvalue for the vector x_i.

Writing Λ for the diagonal matrix of eigenvalues on the right, we write this as

$$MS = S\Lambda.$$

If the eigenvectors are independent, then S is invertible and so we have the following decomposition

$$S^{-1}MS = \Lambda$$

which because Λ is diagonal, we call a *diagonalisation of* M. We also say that S diagonalises M (by conjugation). A fast and easy application of such a diagonalisation is that is allows us to quickly compute M^d.

Example 5.6. Where $M = \left[\begin{smallmatrix} 1 & 2 \\ 0 & 3 \end{smallmatrix}\right]$, one can show that $M = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^{-1}$. So

$$M^5 = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]\left[\begin{smallmatrix} 1 & 0 \\ 0 & 3 \end{smallmatrix}\right]^5\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^{-1} = \left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]\begin{bmatrix} 1^5 & 0 \\ 0 & 3^5 \end{bmatrix}\left[\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right]^{-1}.$$

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The diagonalisation of a matrix, if it exists, need not be unique, Indeed, in our initial example we can re-order the columns of S or multpliy them by constants, and get a different matrix S that also diagonalises M. But as the ith column of MS is λ_i times the ith column of S, the columns of S must be eigenvalues of M.

The multiplicity d of an eigenvalue λ of M as a root of the characteristic polynomial is its *algebraic dimension*. The dimension of its eigenspace is its *geometric dimension*. For M to be diagonalisable, we need that any eigenvalue λ of M of algebraic dimension d has d independent eigenvectors, so has geometric dimension d as well.

Lets observe a couple more properties of eigenvalues. You will be asked to prove simple versions of them.

i. An eigenvector x of M is an eigenvector of M^d (though the converse need not be true) and if its eigenvalue is λ for M then its eigenvalue is λ^d for M^d.

ii. Diagonalisable matrices A and B commute if and only if they are diagonalised by the same matrix S.

Practice

Assuming that M is diagonalisable, show property (i).

Practice

Show property (ii), in the case that all eigenvalues of A are distinct.

Problems from the text

5.2: 1, 4, 8, 11, 14, 20, 23, 27, 28, 31, 44 (3, 7, 5, 13, 12, 16, 21, 25, 27, 32, 45)

5.3 Difference Equations and Powers M^k

We look at two example applications of eigenvalue and eigenvectors to the field of stochastic processes. In the first we see a Markov process, in the second, the stochastic process is described by a difference equation.

5.3.1 Markov Process

In a Stochastic process we have a set of n states $\{1, \dots, n\}$, which an object can be in. The state will change at regular times steps. Where $p_i(t)$ is the probability of being in state i at time $t \in \{0, 1, 2, \dots\}$, the Stochastic process is *Markov* if $p_j(t+1)$ depends entirely on the previous time step, independent of t:

$$p_j(t+1) = t_{1,j} \cdot p_1(t) + t_{2,j} \cdot p_2(t) + \cdots + t_{n,j} \cdot p_n(t),$$

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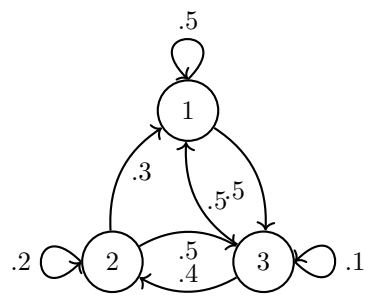
and if for each i the constants $t_{i,j}$ are the probability of moving from state i to state j at a given time step. So in particular, all the $t_{i,j}$ are non-negative and $\sum_{j=1}^{n} t_{i,j} = 1$ for all i.



If at time $t = 0$ there is a $1/3$ probability of being in any state, we can compute the probability if being in state 1 as time $t = 1$ as

$$p_1(1) = 1/3 \cdot t_{1,1} + 1/3 \cdot t_{2,1} + 1/3 \cdot t_{3,1} = 1/3(.5 + .5 + .3) \approx .4333.$$

Our notation is clearly chosen to make a matrix equation out of this. The *state vector*

$$u_t = \left[p_1(t), p_2(t), \ldots, p_n(t) \right]^T$$

describes the states at time t, and where $T = [t_{i,j}]$ is the *transition matrix* we clearly have

$$u_t = Tu_{t-1} = \cdots = T^t u_0.$$

For the Markov process illustrated above we have

$$T = \begin{bmatrix} .5 & .3 & .5 \\ 0 & .2 & .4 \\ .5 & .5 & .1 \end{bmatrix}$$

Such a matrix, one in which the entries non-negative and each columns sums to 1, is called *Markov*. In applications we are interested in what happens to u_t as t changes: does the probabillity of being in a certain state go to 1, and the other states 0? Does the probability of being in a state oscillate? converge? This asymptotic behaviour of the system is related to the eigenvalues of T.

The following is known as the Perron-Frobenius Theorem.

Theorem 5.7. *If T is a Markov matrix then 1 is an eigenvalue, its eigenvector can be taken to have no negative entries, and no other eigenvalue has larger absolute value. Further, if T (or T^k for some k) has no zero entries, then no other eigenvalue has absolute value 1.*

Note

This isn't proved in the book, and we do not prove it here. But he fact that T has a eigenvalue of 1 is easy: as the columns sum to 1, the columns of $T - I$ sum to 0 so are dependent. Thus $\det(T - I) = 0$.

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Using a computer, we have computed the eigenvalues of the matrix T given above:

$$\begin{aligned}
\lambda_1 &= -2/3 & v_1 &= (1, 2, -3) \\
\lambda_2 &= 1/5 & v_2 &= (1, -1, 0) \\
\lambda_3 &= 1 & v_3 &= (1, 5/13, 10/13) = \frac{1}{28}(13, 5, 10)
\end{aligned}$$

We got an eigenvalue of 1 as predicted, and its eigenvector v_3 has all positive entries.

Scaling this eigenvector so that its entries sum to 1, we get a state vector

$$u_t = \frac{1}{28}\begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix}.$$

Applying the transition we get that $u_{t+1} = Tu_t = u_t$ is the same state vector. This state is called a *steady state*. If we start in this state, the system never changes.

But what happens if we start at some other state $u_0 \neq v_3$? To see, we first diagonalise T. With some work one gets:

$$T = S\Lambda S^{-1} = \frac{1}{84}\begin{bmatrix} 1 & 1 & 13 \\ 2 & -2 & 5 \\ -3 & 0 & 10 \end{bmatrix}\begin{bmatrix} -2/3 & & \\ & 1/5 & \\ & & 1 \end{bmatrix}\begin{bmatrix} 10 & 10 & -18 \\ 35 & -49 & -21 \\ 3 & 3 & 3 \end{bmatrix}.$$

Then we can compute $u_t = T^t u_0 = S\Lambda^t S^{-1} u_0$:

$$\begin{aligned}
u_t &= \begin{bmatrix} 1 & 1 & 13 \\ 2 & -2 & 5 \\ -3 & 0 & 10 \end{bmatrix}\begin{bmatrix} -2/3 & & \\ & 1/5 & \\ & & 1 \end{bmatrix}\begin{bmatrix} 5/42 & 5/42 & -3/14 \\ 5/12 & -7/12 & -1/4 \\ 1/28 & 1/28 & 1/28 \end{bmatrix}\begin{bmatrix} p_1 \\ p_2 \\ p_3 \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 & 13 \\ 2 & -2 & 5 \\ -3 & 0 & 10 \end{bmatrix}\begin{bmatrix} -2/3 & & \\ & 1/5 & \\ & & 1 \end{bmatrix}\begin{bmatrix} 5p_1/42+5p_2/42-3p_3/14=:c_1 \\ 5p_1/12-7p_2/12-1p_3/4=:c_2 \\ 1p_1/28+1p_2/28+1p_3/28=:c_3 \end{bmatrix} \\
&= c_1(-2/3)^t\begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix} + c_2(1/5)^t\begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + c_3(1)^t\begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix}
\end{aligned}$$

Now as t gets big, the first two disappear, and so

$$u_t \to c_3\begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix} = \frac{1}{28}\begin{bmatrix} 13 \\ 5 \\ 10 \end{bmatrix},$$

which is our steady state. We don't even have to compute c_3, we know the state vector sums to 0. So the initial position u_0 didn't really matter (as long as it was all non-negative and summed to 1.) **We didn't even have to compute S^{-1}! This will be a good time-save. The positive eigenvector of the eigenvalue 1 is a steady state,** and most of the time (unless there is another eigenvalue of absolute value 1) the system limits to it at t grows.

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Example 5.8. Each year $\frac{1}{10}$ of the people in Daegu move to Ulsan, and $\frac{2}{10}$ of the people in Ulsan move to Daegu. If there are 1000 people in Korea, (and obviously that all live in one of these two cities,) then what are the stable populations of the two cities? (It is not hard to see that $666\frac{2}{3}$ people live in Daegu and $333\frac{1}{3}$ in Ulsan, but... we use matrices.)

Where the state vector $u_t = \begin{bmatrix} D, U \end{bmatrix}$ tells us the respective proportions D and U of the population that live in Daegu and Ulsan, the transition matrix is the Markov matrix

$$T = \begin{bmatrix} .9 & .2 \\ .1 & .8 \end{bmatrix}.$$

All entries are non-zero, so wknow it has only one eigenvalue $\lambda_1 = 1$ with absolute value 1, and so the limiting steady state of the system is the corresponding eigenvector x_1 which is the solution to:

$$0 = \begin{bmatrix} .9-1 & .2 \\ .1 & .8-1 \end{bmatrix} x = \begin{bmatrix} -.1 & .2 \\ .1 & -.2 \end{bmatrix} x.$$

Thus our steady state is $x_1 = (2, 1)$. Scaling so that the total population is 1000, we get $D = 666\frac{2}{3}$ people living in Daegu, and $333\frac{1}{3}$ in Ulsan.

5.3.2 Difference Equations

Consider a recursive description of a population $p(t)$ at time t such as

$$p(0) = 100 \qquad \text{and} \qquad p(t) = p(t-1) \cdot 1.2.$$

We can solve this as

$$p(t) = 100(1.2)^t.$$

This is easy, but it gets more difficult when the recurrence is more complicated.

Example 5.9. The fibonacci recurrence for describing a population of rabbits is

$$F_0 = 0, F_1 = 1, F_2 = 1, \qquad \text{and} \qquad F_{t+2} = F_{t+1} + F_t \text{ for } t \geq 2.$$

To solve this (to find a closed formula for F_t) we look at it as a stochastic process with

$$u_{t+1} = \begin{bmatrix} F_{t+2} \\ F_{t+1} \end{bmatrix} \text{ depending on } u_t = \begin{bmatrix} F_{t+1} \\ F_t \end{bmatrix}.$$

The transmission matrix T should yield

$$T \begin{bmatrix} F_{t+1} \\ F_t \end{bmatrix} = Tu_t = u_{t+1} = \begin{bmatrix} F_{t+2} \\ F_{t+1} \end{bmatrix} = \begin{bmatrix} F_{t+1} + F_t \\ F_{t+1} \end{bmatrix},$$

and so we see that $T = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$.

Now this is not a Markov matrix, so we do not know for sure that there is an eigenvalue of 1. We find the eigenvalues.

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Solving

$$0 = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - \lambda - 1 = (\lambda - 1/2)^2 - 5/4$$

gives $\lambda = \frac{1\pm\sqrt{5}}{2}$. From the second row we get that the eigenvector x_i for λ_i is $(\lambda_i, 1)$. So where $u_0 = (F_1, F_0) = (1, 0)$, we have

$\lambda_1 - \lambda_2 = \frac{1+\sqrt{5}}{2} - \frac{1-\sqrt{5}}{2} = \sqrt{5}$

$$\begin{aligned} u_t &= T^t u_0 = S\Lambda^t S^{-1} u_0 = \left[\begin{smallmatrix} \lambda_1 & \lambda_2 \\ 1 & 1 \end{smallmatrix}\right] \left[\begin{smallmatrix} \lambda_1^t & \\ & \lambda_2^t \end{smallmatrix}\right] \left[\begin{smallmatrix} 1 & -\lambda_2 \\ -1 & \lambda_1 \end{smallmatrix}\right] \frac{(1,0)}{\lambda_1 - \lambda_2} \\ &= \frac{(1, -\lambda_2)\cdot(1,0)}{\sqrt{5}} \lambda_1^t \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} + \frac{(-1, \lambda_1)\cdot(1,0)}{\sqrt{5}} \lambda_2^t \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} \lambda_1^t \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix} - \frac{1}{\sqrt{5}} \lambda_2^t \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{5}} \begin{bmatrix} \lambda_1^{t+1} - \lambda_2^{t+1} \\ \lambda_1^t - \lambda_2^t \end{bmatrix}. \end{aligned}$$

The bottom row of this is

$$F_t = \frac{\lambda_1^t - \lambda_2^t}{\sqrt{5}} = \frac{1}{\sqrt{5}} \left(\left(\frac{1+\sqrt{5}}{2} \right)^t - \left(\frac{1-\sqrt{5}}{2} \right)^t \right) \longrightarrow \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^t$$

as $t \to \infty$.

Note

Actually, the absolute value $\frac{1}{\sqrt{5}} \left| \frac{1-\sqrt{5}}{2} \right|^t$ of the second term is always less than $1/2$, so F_t is just the closest integer to $\frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^t$.

In general when solving a recurrence $u_{t+1} = Tu_t$ for T with distinct eigenvalues, each eigenvalue λ_1 of T yields a part

$$c_1 \lambda_i^t x_i$$

of the solution. The full solution is of the form

$$u_t = c_1 \lambda_1^t x_1 + c_2 \lambda_2^t x_2 + \cdots + c_n \lambda_n^t x_n.$$

Often, the eigenvalues are ordered so that

$$\lambda_n \le \cdots \le \lambda_2 \le \lambda_1.$$

Then λ_1 is the maximum eigenvalue and the difference equation is unstable if $|\lambda_1| > 1$.

As t gets large, the eigenvalues of maximum absolute value dominate, and the other terms can be thrown away.

The difference equation is...

- *stable* if $|\lambda_i| < 1$ for all i, the limiting value is 0.
- *neutrally stable* if $|\lambda_i| = 1$ for some i and $|\lambda_j| < 1$ for all $j \neq i$. The limiting state may be $c_i x_i$, or bounce between $-c_i x_i$ and $c_i x_i$, or if λ_i is complex, it may have recur with some longer period.

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- *unstable* if there is an eigenvalue with $|\lambda_i| > 1$, the state grows infinitly.

Note

What if two eigenvalues have absolute value 1? Is the state not also stable then? Not necessarily, see the text for an example. What if two eigenvalues are 1?

Problems from the text

5.3: 1, 3, 4, 5, 6, 8, 20, 23, 27 (2, 1, 5, 4, 8, 7, 22, 23, 25)

5.4 Differential Equations

A system of differential equations is simply a system of equations involving derivatives. Solving one means finding the functions for which the equations hold.

Example 5.10. Solve the differential equation: $v'(t) = a \cdot v(t)$. Perhaps we remember that $v(t) = e^{at}$ is a solution to this equation, as $\frac{\mathrm{d}}{\mathrm{d}t} e^{at} = ae^{at}$.

As $v(t) = ce^{at}$ is also a solution, there is usually and extra 'initial value' needed to determine a particular solution.

Practice

Find $v(t)$ when $v'(t) = av(t)$ and $v(0) = 4$.

This becomes more difficult with more variables.

Example 5.11. Solve

$$\begin{aligned} \frac{\mathrm{d}v}{\mathrm{d}t} &= 4v - 5w \text{ where} \qquad v(0) = 8 \\ \frac{\mathrm{d}w}{\mathrm{d}t} &= 2v - 3w \text{ where} \qquad w(0) = 5 \end{aligned}$$

To solve this, we will first guess that the solution is of the form

$$v(t) = ae^{\lambda t} \qquad w(t) = be^{\lambda t}$$

for some constants a, b and λ, and then find these constants.

Plugging our guesses into the given equations, we get

$$\begin{aligned} \lambda ae^{\lambda t} &= 4ae^{\lambda t} - 5be^{\lambda t} \\ \lambda be^{\lambda t} &= 2ae^{\lambda t} - 3be^{\lambda t}. \end{aligned}$$

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Defining the matrices

$$u(t) = \begin{bmatrix} v(t) \\ w(t) \end{bmatrix} \qquad T = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}$$

this the orignal system can be written as

$$u' = \frac{\mathrm{d}}{\mathrm{d}t} u = Tu \qquad u(0) = (8, 5) \tag{4}$$

where $u' = \frac{\mathrm{d}}{\mathrm{d}t} u$ is the obvious shorthand for $(v'(t), w'(t))$. With our guessed solution plugged in, this becomes

$$e^{\lambda t} \lambda \begin{bmatrix} a \\ b \end{bmatrix} = e^{\lambda t} T \begin{bmatrix} a \\ b \end{bmatrix} \tag{5}$$

which has the same solutions as $T[\begin{smallmatrix} a \\ b \end{smallmatrix}] = \lambda[\begin{smallmatrix} a \\ b \end{smallmatrix}]$. So to find a, b and λ we have to find the eigens of T.

You can do this:

$$\begin{aligned} \lambda_1 &= -1 & x_1 &= (1, 1) \\ \lambda_2 &= 2 & x_2 &= (5, 2) \end{aligned}$$

Each pair (λ_i, x_i) gives a solutions to the system $u' = Tu$, so we have two:

$$\begin{aligned} u(t) &= e^{\lambda_1 t} x_1 = e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ u(t) &= e^{\lambda_2 t} x_2 = e^{2t} \begin{bmatrix} 5 \\ 2 \end{bmatrix} \end{aligned}$$

and as a linear combination of solutions is a solution, we have a solution space

$$u(t) = c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} 5 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} e^{-t} & \\ & e^{2t} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}.$$

Our initial conditions $u(0) = (5, 8)$ designated one solution:

$$\begin{bmatrix} 8 \\ 5 \end{bmatrix} = u(0) = \underbrace{\begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix}}_{S} \underbrace{\begin{bmatrix} e^0 & \\ & e^0 \end{bmatrix}}_{e^{\Lambda t}} \underbrace{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}}_{S^{-1}u(0)} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

which we can solve:

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 & 5 \\ 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 8 \\ 5 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} 2 & -1 \\ -5 & 1 \end{bmatrix} \begin{bmatrix} 8 \\ 5 \end{bmatrix} = \frac{1}{-3} \begin{bmatrix} -9 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Thus $u(t) = 3e^{-t}[\begin{smallmatrix} 1 \\ 1 \end{smallmatrix}] + e^{2t}[\begin{smallmatrix} 5 \\ 2 \end{smallmatrix}]$ is our solution.

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Note

Summarising: the differential equation $\frac{\mathrm{d}}{\mathrm{d}t}u = Tu$ has the solution

$$u(t) = Se^{\Lambda t}S^{-1}u(0)$$

where T is diagonalised $T = S\Lambda S^{-1}$.

5.4.1 Matrix exponential

We cheated with the notation a bit above, writing

$$e^{\Lambda t} = \begin{bmatrix} e^{\lambda_1 t} & & \\ & e^{\lambda_2 t} & \\ & & e^{\lambda_d t} \end{bmatrix}.$$

Lets define this properly. Recall that the McClauren series for e^x is

$$e^x = 1 + x + x^2/2! + x^3/3! + \dots$$

In analogue to this we defined, for a matrix M,

$$e^{Mt} = I + Mt + (Mt)^2/2! + (Mt)^3/3! + \dots.$$

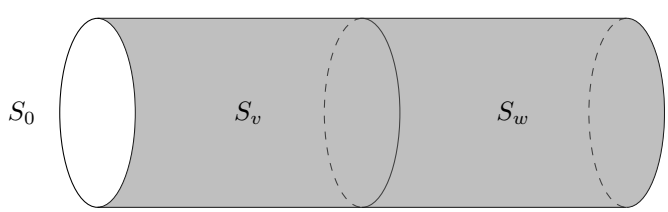
This agrees with our earlier use of the notation, and also has some properties that are envoked by the notation:

- $e^{Ms}e^{Mt} = e^{M(s+t)}$
- $e^{Mt}e^{-Mt} = I$
- $\frac{\mathrm{d}}{\mathrm{d}t}e^{Mt} = Me^{Mt}$.

Again, in this last, the notation $\frac{\mathrm{d}}{\mathrm{d}t}M$ means that we take a derivative entrywise. Thus e^{Mt} is the solution to $u'(t) = Mu(t)$, and in fact $e^{Mt} = Se^{\Lambda t}S^{-1}$.

5.4.2 A physical example

Differential equations can arise from looking at the movement of gas in a pipe.



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In two regions S_v and S_w, we start at time $t = 0$ with concentrations $v(0)$ and $w(0)$. In the region S_0 outside the pipe, we assume the concentration of gas is essentially 0. At time t gas moves between regions proportionally to the difference in concentrations. So the flow is

$$\frac{\mathrm{d}v}{\mathrm{d}t} = (0 - v) + (w - v) \qquad \frac{\mathrm{d}w}{\mathrm{d}t} = (0 - w) + (v - w).$$

To find $v(t)$ and $w(t)$ we solve $u' = \left[\begin{smallmatrix} -2 & 1 \\ 1 & -2 \end{smallmatrix}\right] u$. We start be diagonalising the transition matrix $T = \left[\begin{smallmatrix} -2 & 1 \\ 1 & -2 \end{smallmatrix}\right]$:

$$T = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 & \\ & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1}$$

$$\begin{aligned} u(t) &= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} e^{-t} & \\ & e^{-3t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}^{-1} u(0) \\ &= c_1 e^{-t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-3t} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \end{aligned}$$

where $\left[\begin{smallmatrix} c_1 \\ c_2 \end{smallmatrix}\right] = \left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]^{-1} u(0)$.

5.4.3 Stability

For the equation $u' = Tu$ the solution is of the form

$$u(t) = c_1 e^{\lambda_1 t} x_1 + c_2 e^{\lambda_2 t} x_2 + \cdots + c_n e^{\lambda_n t} x_n$$

where $T^{-1}u(0) = [c_1, \ldots, c_n]^T$. This is stable if $e^{\lambda_i t} \to 0$ for all i and (usually) blows up if any of the $e^{\lambda_i t}$ go to infinity. Observing that

$$|e^{\lambda t}| = e^{at} \text{ where } \lambda = a + bi$$

we see that $e^{\lambda_i t}$ goes to infinity iff the real part $Re(\lambda)$ of λ is greater than 0.

Note

The system $u' = Tu$ is...

- *stable* if $Re(\lambda_i) < 0$ for all i.
- *neutrally stable* if $Re(\lambda_i) \leq 0$ for all i and $Re(\lambda_j) = 0$ for some j.
- *unstable* if $Re(\lambda_i) > 0$ for some i.

Problems from the text

5.4: 1, 2, 3, 6, 7, 10, 16, 18 (1, 2, 3, 9, 8, 5, 20, 19)

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5.5 Complex Matrices

Okay, complex numbers are sneaking into our examples. The picture is incomplete without them. Let's look at matrices over the field $\mathbb{C}$ of complex numbers. This requires us to generalise several definitions such as length and inner (dot) product.

5.5.1 Complex Numbers

Recal lthat complex numbers are of the form

$$z = a + ib$$

where $a, b \in \mathbb{R}$ and $i^2 = -1$. So arithmetic works as expected:

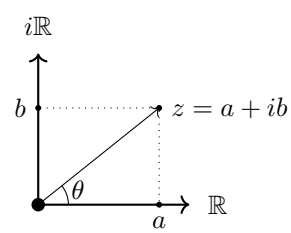
- $(a_1 + ib_1) + (a_2 + ib_2) = (a_1 + a_2) + i(b_1 + b_2)$, and
- $(a_1 + ib_1)(a_2 + ib_2) = (a_1a_2 - b_1b_2) + i(a_1b_2 + a_2b_1)$.

For a complex number $z = a + ib$, the value $a = Re(z)$ is the *real part* of z and $b = Im(z)$ is the *imaginary part*.

Multiplication is easier to deal with if we use polar notation. Recall, in polar notation, that the complex number $z = a + ib$ can be written as

$$re^{i\theta} = r\sin\theta + ir\cos\theta$$

where $r = \sqrt{a^2 + b^2}$ is length of z and $\theta = \arctan(b/a)$ (for $a \geq 0$) is the angle that z makes with the real axis when we graph it in the complex plane:



Now, multiplication works as:

$$r_1e^{i\theta_1} \cdot r_2e^{i\theta_2} = (r_1 \cdot r_2)e^{i(\theta_1+\theta_2)}.$$

Moreover, this length r, this is what we use as the length $|z|$ of the z. A complex number $z = a+ib$ is real if $b = 0$, and in this case $|z| = \sqrt{a^2 + b^2} = \sqrt{a^2} = |a|$, so this generalises our notion of absolute value for real numbers.

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5.5.2 Complex vectors: Length and inner product

Let $x = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}$ be in $\mathbb{C}^n$. The *length* or *norm* $\|x\|$ is defined by

$$\|x\|^2 = |z_1|^2 + \cdots + |z_n|^2.$$

Practice

What is the length of the complex vector $x = (2, 1 + 2i)$?

The inner product for real vectors was

$$x \cdot y = x^T y = x_1 \cdot y_1 + \cdots + x_n \cdot y_n.$$

We could use this same formula for complex numbers, but then we would loose such properties as

$$x \cdot x = \|x\|^2.$$

Practice

Give an example showing this.

To define a proper inner product, we first need to define a 'complex conjugate'. For $z = a + ib = re^{i\theta}$, the *complex conjugate* of z is the number

$$\bar{z} = a - ib = re^{-i\theta}.$$

Practice

Show that $z\bar{z} = |z|^2$.

Now, for complex (column) vectors $x = (x_1, \ldots, x_n)^T$ and $y = (y_1, \ldots, y_n)^T$ we let $\bar{x} = (\bar{x}_1, \ldots, \bar{x}_n)^T$ and define the inner product as

$$x \cdot y = \bar{x}^T y = \bar{x}_1 \cdot y_1 + \cdots + \bar{x}_n \cdot y_n.$$

It follows then that

$$\begin{aligned} x \cdot x = \bar{x}^T x &= \bar{x}_1 \cdot x_1 + \cdots + \bar{x}_n \cdot x_n \\ &= |x_1|^2 + \cdots + |x_n|^2 = \|x\|. \end{aligned}$$

5.5.3 The Hermitian

The *(complex) conjugate* of a matrix $M = [m_{ij}]$ is the matrix

$$\bar{M} = [\bar{m}_{ij}]$$

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we get from M by entrywise conjugation.

So, for example

$$\overline{\begin{bmatrix} 1 & 4 \\ 1+i & i \end{bmatrix}} = \begin{bmatrix} 1 & 4 \\ 1-i & -i \end{bmatrix}.$$

The *Hermitian* or *conjugate transpose* of M is

$$M^H := \bar{M}^T = \bar{M^T}.$$

Practice

What is the Hermitian of $\left[\begin{smallmatrix} 2+i & 4 \\ i & 0 \\ 1 & 2-i \end{smallmatrix}\right]$?

Observe that

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}^H = [\bar{x}_1, \ldots, \bar{x}_n],$$

so the inner product can now be written as $x \cdot y = x^H y$.

Though $x^T = x^H$ when x is a real vector, so $x^T x = x^H x$, this is not true for complex vectors x. In many situations, when we are considering complex vectors, we must use x^H instead of x^T in results or definition we know from the real case. One such situation is the definition of orthogonality. From now on:

Note

Two vectors x and y are *orthogonal* if $x^H y = 0$.

Observe also, that much like we had for the transpose, we have

$$(AB)^H = (\bar{AB})^T = (\bar{B^T}\bar{A^T}) = \bar{B^T}\ \bar{A^T} = B^H A^H.$$

Practice

Why does that second last equality hold?

A matrix M is *Hermitian* if

$$M^H = M.$$

For example $\left[\begin{smallmatrix} 2 & 3-i \\ 3+i & 5 \end{smallmatrix}\right]$ is Hermitian. One quickly sees that for a Hermitian matrix $M = [m_{ij}]$ we must have

- m_{ii} is real, and
- $\bar{m}_{ij} = m_{ji}$.

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Hermitian matrices have three nice properties, and keep in mind, **if a matrix is real, it is Hermitian if and only if it is symmetric,** and so these properties hold for real symmetric matrices too. Let M be Hermitian.

i. For all $x \in \mathbb{C}^n$, $x^H M x$ is real.

ii. All eigenvalues of M are real.

iii. Eigenvectors of M with distinct eigenvalues are orthogonal.

As a hermitian matrix M has distinct eigenvalues, we can diagonalise it as

$$M = S\Lambda S^{-1}.$$

The orthogonality of the eigenvectors means that if normalised, then $SS^H = I$, and so we get

$$M = S\Lambda S^H,$$

which is nice as S^H is much easier to compute than S^{-1} is in general.

Note

We will show in the next section that this result actually holds for any Hermitian matrix M, not only those with distinct eigenvalues. This will come down to showing that for a Hermitian matrix, every eigenvalue has the same geometric mutliplicity as algebraic multiplicity.

Proof of properties i. ii. and iii.

To see property i., observe that the 1×1 matrix $x^H M x$ is its own conjugate:

$$(x^H M x)^H = x^H M^H x^{HH} = x^H M^H x = x^H M x$$

and so must be real.

To see property ii., let $Mx = \lambda x$. Then

$$\underbrace{x^H M x}_{\text{real by i.}} = x^H \lambda x = \lambda x^H x = \lambda \|x\|$$

and $\|x\|$ is real (and non-zero) so $\lambda = \frac{x^H M x}{\|x\|}$ is real.

To see property iii., let

$$Mx = \lambda_1 x \text{ and } My = \lambda_2 y.$$

Then

$$\lambda_1 x^H y = (Mx)^H y = x^H M y = x^H \lambda_2 y = \lambda_2 x^H y,$$

so $\lambda_1 \neq \lambda_2$ gives that $x^H y = 0$, as needed.

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5.5.4 Unitary Matrices

Generalising on the idea of an orthogonal matrix, a complex matrix U is *unitary* if

$$U^H U = I.$$

Practice

Show the following properties for any unitary matrix U.

i. $(Ux)^H(Uy) = x^H y$ and in particular $\|Ux\| = \|x\|$.

ii. Every eigenvalue λ of U has $|\lambda| = 1$.

iii. Eigenvectors of different eigenvalues are orthonormal.

As we mentioned above, we will show in the next section that every Hermitian matrix will be diagonalisable by a unitary matrix. Let's see an example of this.

Example 5.12. The matrix $M = \left[\begin{smallmatrix} 3 & 2+2i \\ 2-2i & 1 \end{smallmatrix}\right]$ is Hermitian. Let's diagonalise it.

$$\begin{aligned} Det(M - \lambda I) &= (3-\lambda)(1-\lambda) - (2+2i)(2-2i) = (3-\lambda)(1-\lambda) - 8 \\ &= \lambda^2 - 4\lambda - 5 = (\lambda - 2)^2 - 9 \end{aligned}$$

has roots $\lambda = 2 \pm \sqrt{9} = -1, 5$. To get the eigenvector for $\lambda_1 = 5$ we find the null-vector of $\left[\begin{smallmatrix} -2 & 2+2i \\ 2-2i & -4 \end{smallmatrix}\right]$ which is $x_1 = (1+i, 1)$, and read off that the eigenvector of λ_2 is $x_2 = (1+i, -2)$. These should be orthogonal:

$$(1-i, 1) \cdot (1+i, -2) = (1-i)(1+i) + (1)(-2) = 2 - 2 = 0.$$

Normalising them to $x_1 = (1+i, 1)/\sqrt{3}$ and $x_2 = (1+i, -2)/\sqrt{6} = (1, i-1)/\sqrt{3}$ we get that $U = \frac{1}{\sqrt{3}}\left[\begin{smallmatrix} 1+i & 1 \\ 1 & -1+i \end{smallmatrix}\right]$ and so $U^{-1} = U^H = \frac{1}{\sqrt{3}}\left[\begin{smallmatrix} 1-i & 1 \\ 1 & -1-i \end{smallmatrix}\right]$.

We should check that $M = U \begin{bmatrix} -1 & \\ & 5 \end{bmatrix} U^{-1}$.

Problems from the text

5.5: 1, 2, 6, 10, 12, 15, 18, 29, 37, 42, 43, 49, 50 (1, 7, 2, 6, 18, 17, 12, 30, 40, 37, 50, 41, 36)

5.6 Similarity Transformations

A matrix B is similar to a matrix A if

$$B = MAM^{-1}$$

for some matrix M.

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Similar matrics share similar properties For example, if $B = MAM^{-1}$ then A and B have the same eigenvalues, and their eigenvectors are related through M:

$$\begin{aligned} Ax = \lambda x \quad &\iff \quad M^{-1}BMx = \lambda x \\ &\iff \quad B(Mx) = \lambda(Mx) \end{aligned}$$

Practice

Show that if B is similar to a diagonalisable matrix, then B is diagonalisable.

Indeed, that a matrix is diagonalisable just means that it is similar to a diagonal matrix. Recalling our long goal of computing the eigenvalues of a big matrix, this suggests where we might be going. If we can find a diagonalisation of a matrix M then we have its eigenvalues. For this goal, it enough that M is similar to a triangular matrix. But we will see in the next subsection why it is nicer if M is similar to a diagonal matrix. We will see also that while similarity is nice, similary by unitary matrices is nicer. So the ideal will be writing $M = U\Lambda U^{-1}$ for some unitary U and diagonal Λ. We will look at which matrices we can do this for. It won't be all. For the other matrices we will show that they can be triangulised, but better than this, triangulised by unitary matrices. And that they can also by 'almost diagonalised', into something we will call theIR Jordan form.

First, we look, from a new point of view, at why diagonalising is nicer than triangulating, if we can do it.

5.6.1 Change of Basis

Given a basis $B = \{b_1, \ldots, b_n\}$ and a vector x the notation

$$[x]_B = (a_1, \ldots, a_n)$$

tells us that $x = a_1b_1 + a_2b_2 + \cdots + a_nb_n$.

Example 5.13. Where $E = \{e_1, e_2\}$ is the standard basis $[(1,3)]_E = (1,3)$ but where B is the basis $B = \{b_1 = (1,1), b_2 = (-1,1)\}$ we have $[(1,3)]_B = (2,1)$ because $(1,3) = 2(1,1) + 1(-1,1)$.

Given a vector $[x]_B$ represented in the basis B, it is easy to transform it to the standard basis E, but how do we find $[x]_B$ for a given vector x, as we did in the above example? Well, the vector $b_1 = \left[\begin{smallmatrix}1\\1\end{smallmatrix}\right]$ is expressed in basis B as $[b_1]_B = \left[\begin{smallmatrix}1\\0\end{smallmatrix}\right]$. This can be written as

$$\begin{bmatrix}1\\1\end{bmatrix} = M_B \begin{bmatrix}1\\0\end{bmatrix}$$

where M_B is the matrix whose i^{th} column is the ith vector b_i in B. More generally we have that if $[x]_b = (a_1, \ldots, a_n)$, so $x = a_1b_1 + a_2b_2 + \cdots + a_nb_n$, then

$$x = M_B[x]_B.$$

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So to go in the other direction, to find the representation $[x]_B$ of the vector x in the basis B, we compute

$$[x]_B = M_B^{-1}x.$$

Example 5.14. To represent $x = (1,3)$ in the basis $B = \{(1,1),(-1,1)\}$, we find $M_B^{-1} = \left[\begin{smallmatrix} 1 & 1 \\ -1 & 1 \end{smallmatrix}\right]/2$ and compute

$$[(1,3)]_B = M_B^{-1}x = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Note:

Change of basis can be confusing. We talk of the vectors x and $M_B^{-1}x$ as being the same vector, just represented in different bases. But hey, multiplying by the transformation M_B^{-1} takes the vector $(1,3)$ to $(2,1)$! It does change. Yep, confusing. The point of view of the change of basis is that we are keeping the vectors still, but moving the paper (and the coordinate axes). Instead of rotating vectors clockwise, we rotate the paper counterclockwise. Instead of stretching the vectors, we zoom out the paper. Multiplying by M_B changes from the B frame of reference to the standard E frame of reference. (Notation can be the other way around in other books).

Example 5.15. The matrix $A = \left[\begin{smallmatrix} 3 & 1 \\ 1 & 3 \end{smallmatrix}\right]$ diagonalises to

$$A = M\Lambda M^{-1} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 4 & \\ & 2 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}/2.$$

Applying this as a transformation to some vector $x = (1,3)$ we get

$$A : (1,3) \xmapsto{M_B^{-1}} (2,1) \xmapsto{\Lambda} (8,2) \xmapsto{M_B} (6,10).$$

We interpret transformation by A as consisting of three steps. The first is changing $x = (1,3)$ to $[x]_B = (2,1)$ in the basis B. Then in the basis B transforming by Λ to $(8,2)$. Transformations by diagonal matrices are easy to visualise as there is no rotation involved, just stretching in the two directions. Finally, we change back to $(6,10)$ in the original basis.

So a matrix A being similar to a diagonal matrix is pretty nice. It is even nicer if the matrix M that transforms A to diagonal is unitary (orthanormal). Not only is M^{-1} then easy to compute, but if M is unitary, then the eigenspaces of A are orthogonal, so basis we are transforming to is orthonormal. Thus the action of A decomposes into projections onto the eigenspaces:

$$A = U\Lambda U^H = \lambda_1 x_1 x_1^H + \cdots + \lambda_n x_n x_n^H.$$

Example 5.16. Where

$$\begin{aligned} A = \frac{1}{2}\begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} &= \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 2 & & \\ & 1 & \\ & & 1 \end{bmatrix}\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} & 0 \\ 1/\sqrt{2} & -1/\sqrt{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= 2\begin{bmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 0 \end{bmatrix} + 1\begin{bmatrix} 1/2 & -1/2 & \\ -1/2 & 1/2 & \\ & & \end{bmatrix} + 1\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

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The first matrix on the left of the last line is the projection onto the eigenspace $\langle(1,1,0)\rangle$ of $\lambda_1 = 2$ while the second two matrices are the projection onto the eigenspace $\langle(1,-1,0),(0,0,1)\rangle$ of $\lambda_2 = 1$.

5.6.2 Triangulating by Unitary Matrices

We cannot always diagonalise a matrix by a unitary matrix, but we can at least triangulate it. The following is called Schur's Lemma.

Theorem 5.17. *Any matrix A can be triangulated: $A = UTU^H$ by a unitary matrix U. The eigenvalues of A are on the diagonal of T.*

Proof. If $A = UTU^H$ for triangular T the fact that the diagonals of T are the eigenvalues of A is immediate from the fact that similar matrices share eigenvalues. So it is enough to prove the first statement of the theorem. See the text for this. □

With this we can now answer the question of which matrices A can be diagonalised by unitary matrices. A matrix A is *normal* if $A^H A = AA^H$.

Practice

Show that $(A^H A)^H = A^H A$ for any matrix A.

Corollary 5.18. *The following are equivalent.*

i. $A = U\Lambda U^H$ *for some unitary matrix* U.

ii. A *has* n *orthogonal eigenvectors.*

iii. A *is normal.*

Proof. The proof of *i.* $\iff$ *ii.* is trivial as the columns of such a U must be eigenvectors. The equivalence of *ii.* and *iii.* comes from observing that where $A = UTU^H$ we have

$$AA^H = UTT^HU^H \text{ and } A^HA = UT^HTU^H$$

and so

$$(AA^H = A^HA) \iff (TT^H = T^HT).$$

Showing that $TT^H = T^HT$ for triangular T if and only if T is diagonal then yields the result. Certainly if T is diagonal we have $TT^H = T^HT$. Question 21 of the text asks you to show that if $TT^H = T^HT$ for triangular T, then T is diagonal. □

Observe now that hermitian matrices A are normal, so whether or not their eigenvalues are distinct, the are diagonalisable by unitary matrices. We claimed this last section without proof.

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5.6.3 Jordon Canonical Form

If we relax the restriction to conjugation by unitary matrices, we can show that a matrix is similar to something even closer to diagonal that a triangular matrix.

An $n \times n$ $M = [m_{ij}]$ matrix is *block diagonal* if the index set $[n]$ can be partitioned into sets $B_1, \ldots, B_d$ of consecutive indices such that m_{ij} is non-zero only if i and j are in the same set B_k for some k. The square submatrices made of the rows and columns in B_k for each k are called *blocks*. A block is *Jordan* if its diagonal entries are all the same, and the only other non-zero entries are ones on the super-diagonal. A matrix is *Jordan* if it is block diagonal with Jordan blocks.

The proof of the following theorem can be found in the appendix of the text.

Theorem 5.19 (Jordon Form)**.** *Any matrix A can be written (uniquely) as $A = MJM^{-1}$ for a Jordon matrix J. (The eigenvalues of A are the diagonals of J. For this representation to be unique, the diagonals must assumed to be non-decreasing, and for Jordan block with the same diagonals, the small block must be assumed to come first.)*

If $A = MJM^{-1}$ for a jordan matrix J, J is the *Jordan canonical form* of A.

Example 5.20. The matrices $A = \begin{bmatrix} 11 & -2 & 7 \\ 0 & 8 & 0 \\ 3 & -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 9 & -2 & -5 \\ 4 & 8 & 4 \\ 1 & -2 & 3 \end{bmatrix}$ both have eigenvalues $4, 8, 8$. For A the eigenspacess have dimensions 1 and 2, for B they have dimensions 1 and 1. So the Jordon forms are

$$J_A = \begin{bmatrix} 4 & & \\ & 8 & \\ & & 8 \end{bmatrix} \qquad \text{and} \qquad J_B = \begin{bmatrix} 4 & & \\ & 8 & 1 \\ & & 8 \end{bmatrix}$$

respectively.

There may be more than one Jordon block for a given eigenvalue, as we see with A above. The number of Jordon blocks for an eigenvalue is determined by the algebraic multiplicity and the geometric multiplicity of that eigenvalue. Knowing that there are two blocks for an eigenvalue does not tell us the size of each.

The following is an easy corollary of the above theorem.

Practice

Show that two matrices are similar if and only if they have the same Jordon Canonical Form.

Not only is the Jordon form useful in that it lets us decide if two matrices are similar, as with a diagonalisation, it is useful for taking powers.

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Practice

Show that

$$\begin{bmatrix} a & & \\ & b & 1 \\ & & b \end{bmatrix}^n = \begin{bmatrix} a^n & & \\ & b^n & nb^{n-1} \\ & & b^n \end{bmatrix}$$

and that

$$\begin{bmatrix} a & 1 & \\ & a & 1 \\ & & a \end{bmatrix}^n = \begin{bmatrix} a^n & na^{n-1} & \binom{n}{2}a^{n-1} \\ & a^n & na^{n-1} \\ & & a^n \end{bmatrix}.$$

Problems from the text

5.6: 1, 3, 5, 6, 8, 11, 14, 16, 26, 36, 40 (5, 2, 7, 4, 8, 11, 18, 13, 30, 38, 42)

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References

[1] Gilbert Strang *Linear Algebra and its applications* Fourth Edition, International Student Edition. 2006 Thompson Learning.

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